



**UNIVERSITY OF CHENAB**



# ***SOFTWARE PROJECT MANAGEMENT***

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Introduction

3DPanoSphere is a mobile application designed to provide users with an interactive platform for creating, sharing, and exploring 360° panoramic images and videos. The app enables users to upload panoramic content, add interactive hotspots, create virtual tours, and navigate through panoramic stories. Social engagement is integrated through likes, comments, following, real-time messaging, and a star rating system to evaluate content quality. The platform focuses on immersive experiences for tourism, education, real estate, and other domains where 360° visuals enhance user understanding and interaction.

Step 1: Select Project & Determine Scope

Activity 1: Identify Candidate Projects from Strategic Plans

Strategic Need Identified

The 3DPanoSphere project was identified from the observation that industries such as real estate, tourism, and education lack an effective and user-friendly platform for sharing immersive 360° virtual experiences. The growing demand for digital transformation and remote interaction made this a high-priority initiative.

How This Project Was Selected

The project was selected based on the following strategic factors:

- ◆ Growing market demand for virtual and immersive content across multiple industries
- ◆ Gap in the market: existing platforms are either costly, limited in features, or lack interactivity
- ◆ University-level project aligned with the BSSE program's goal of building real-world software solutions
- ◆ Opportunity to develop expertise in mobile development, 360° media processing, and social platforms
- ◆ Scalable idea with potential for future commercialization

Project Summary

3DPanoSphere is a mobile application that allows users to upload, manage, and explore 360° panoramic images and videos. It supports the creation of virtual tours with multiple scenes, interactive hotspots, and social features such as likes, comments, messaging, and a star rating system.

Activity 2: Evaluate Projects Against Business Objectives

This activity checks how well the project aligns with the strategic and business goals of the organization.

Strategic Alignment

| Business Objective                     | How 3DPanoSphere Aligns                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------|
| <i>Leverage emerging technologies</i>  | Uses 360° panoramic technology and real-time features to enhance digital user experience        |
| <i>Increase online engagement</i>      | Social features (likes, comments, follows, messaging) drive active user participation           |
| <i>Support industry digitalization</i> | Serves real estate, tourism, and education businesses with virtual property and campus tours    |
| <i>Build scalable solutions</i>        | Modular mobile architecture supports future enhancements like AI recommendations and VR support |

|                                   |                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------|
| <i>Develop in-house expertise</i> | Builds developer skills in panoramic image processing, Spring Boot APIs, and Flutter mobile development |
|-----------------------------------|---------------------------------------------------------------------------------------------------------|

Stakeholder Benefit Summary

| Stakeholder                   | Expected Benefit                                                          |
|-------------------------------|---------------------------------------------------------------------------|
| <i>Real Estate Businesses</i> | Virtual property tours; reduce need for physical visits                   |
| <i>Tourism Industry</i>       | Destination previews; immersive travel planning                           |
| <i>Education Institutions</i> | Virtual campus tours; remote field trips                                  |
| <i>Content Creators</i>       | Portfolio showcase; community engagement; monetization potential          |
| <i>General Users</i>          | Explore real-world locations; social discovery; interactive entertainment |

Activity 3: Conduct Initial Cost-Benefit Analysis

Estimated Project Cost

| Cost Category         | Item                                                        | Estimated Cost (USD)  |
|-----------------------|-------------------------------------------------------------|-----------------------|
| <i>Infrastructure</i> | Cloud hosting (backend server) — \$20/month × 12            | \$240                 |
| <i>Infrastructure</i> | Cloud storage (500GB for panoramic media) — \$30/month × 12 | \$360                 |
| <i>Infrastructure</i> | Domain registration                                         | \$15                  |
| <i>Equipment</i>      | 360° camera for testing (optional; smartphone usable)       | \$300                 |
| <i>Deployment</i>     | Google Play Store developer account (one-time)              | \$25                  |
| <i>Deployment</i>     | Apple App Store developer account (annual)                  | \$99                  |
| <i>Miscellaneous</i>  | Documentation, printing, academic submission                | \$50                  |
| <i>Contingency</i>    | 15% reserve for unexpected expenses                         | \$346                 |
| <i>TOTAL</i>          |                                                             | \$2,680 (PKR 742,000) |

Estimated Benefits

| Benefit Type                  | Description                                                   | Estimated Value (USD) |
|-------------------------------|---------------------------------------------------------------|-----------------------|
| <i>Cost Savings</i>           | Avoiding costs of existing platforms and physical visits      | \$2,000               |
| <i>Revenue Potential</i>      | Future monetization through premium features and partnerships | \$0                   |
| <i>Productivity Gains</i>     | Efficiency improvements for content creators and businesses   | \$500                 |
| <i>Compliance Value</i>       | Data privacy and content standards adherence                  | \$500                 |
| <i>Decision Support</i>       | Analytics tools for creators to improve content strategy      | \$500                 |
| <i>Reduced Maintenance</i>    | Modular architecture reduces long-term maintenance costs      | \$1,000               |
| <i>Avoided Licensing Fees</i> | No ongoing fees from competitor platforms                     | \$2,000               |
| <i>TOTAL BENEFIT</i>          |                                                               | \$6,500               |

Technical Feasibility

The technical feasibility study evaluates whether the system can be built with available technologies, tools, and skills.

| Feasibility Area             | Assessment                                                                             | Conclusion         |
|------------------------------|----------------------------------------------------------------------------------------|--------------------|
| <i>Mobile Development</i>    | Flutter SDK supports Android and iOS development from a single codebase                | Feasible           |
| <i>Backend &amp; APIs</i>    | Spring Boot (Java) is a mature, widely-used backend framework                          | Feasible           |
| <i>360° Panorama Viewing</i> | Established panorama libraries (e.g., Panorama360, custom WebGL viewers) are available | Feasible           |
| <i>Database</i>              | MS SQL Server Developer Edition is free; cloud alternatives available                  | Feasible           |
| <i>Cloud Storage</i>         | AWS S3 / Firebase Storage support large media files at affordable rates                | Feasible           |
| <i>Real-Time Messaging</i>   | WebSocket or Firebase Realtime Database enables live chat features                     | Feasible           |
| <i>AI Recommendations</i>    | Deferred to v2.0; not required in MVP                                                  | Partially Deferred |

| Feasibility Area | Assessment                                                            | Conclusion |
|------------------|-----------------------------------------------------------------------|------------|
| <i>Security</i>  | JWT authentication, encrypted storage, and HTTPS ensure data security | Feasible   |

Financial Feasibility

The financial feasibility confirms that the total project cost of \$2,680 is affordable for an academic project. Development labor is self-funded (zero cost), open-source tools are used, and cloud infrastructure costs are kept minimal by using basic-tier plans.

| Financial Factor                         | Status                                                 |
|------------------------------------------|--------------------------------------------------------|
| <i>Development Labor Cost</i>            | PKR 0 (self-developed as academic project)             |
| <i>Software Tools</i>                    | PKR 0 (all open-source: Flutter, Spring Boot, VS Code) |
| <i>Annual Cloud &amp; Infrastructure</i> | ~\$615/year (hosting, storage, domain)                 |
| <i>One-time Setup Costs</i>              | ~\$424 (camera optional, app store accounts)           |
| <i>Contingency Budget</i>                | \$346 (15% reserve)                                    |
| <i>Funding Source</i>                    | Personal/academic funding                              |
| <i>Break-even Potential</i>              | Achievable within the first year of commercialization  |

Operational Feasibility

The application is designed to be user-friendly and accessible on standard smartphones. No specialized hardware is required for end-users. The admin panel allows easy content moderation, and the system is designed to operate with minimal ongoing maintenance.

- ◆ Target users (real estate agents, tourists, students, general public) can adopt the app with minimal training
- ◆ Onboarding tour is built into the app to guide new users
- ◆ Backend infrastructure is manageable by a single developer during the academic phase
- ◆ Scalable architecture allows future growth without full redesign

Schedule Feasibility

| Phase                        | Duration | Key Activities                                              |
|------------------------------|----------|-------------------------------------------------------------|
| <i>Requirement Gathering</i> | 2 Weeks  | Functional/non-functional requirements, feasibility studies |
| <i>System Design</i>         | 2 Weeks  | Architecture, database design, UI/UX mockups                |
| <i>Development</i>           | 4 Weeks  | 11 modules implemented (frontend + backend)                 |
| <i>Testing</i>               | 1 Week   | Unit, integration, and user acceptance testing              |
| <i>Deployment</i>            | 1 Week   | App store submission, cloud setup, documentation            |



Legal Feasibility

- ◆ User data is protected under applicable data privacy standards
- ◆ Content moderation tools are built in to handle copyright infringement and inappropriate content
- ◆ Users must accept Terms & Conditions and Privacy Policy before using the platform
- ◆ App store compliance guidelines (Google Play and Apple App Store) will be met before deployment

Social Feasibility

The platform addresses a genuine social need for accessible virtual exploration. It democratizes immersive experiences for users who cannot physically visit locations. Industries such as education and tourism benefit from enhanced digital reach. No significant social risks are anticipated beyond standard content moderation needs.

Activity 5: Obtain Management Approval to Proceed

Key Reasons for Approval:

- ◆ The project is technically feasible using available, mature technologies (Flutter, Spring Boot, MS SQL Server)
- ◆ The total estimated cost of \$2,680 is affordable and well within the academic project budget
- ◆ The cost-benefit analysis shows a net benefit of approximately \$65,00 making it highly worthwhile
- ◆ The project aligns with digital transformation trends in real estate, tourism, and education
- ◆ All key stakeholders show positive impact ratings
- ◆ Risk mitigation strategies are in place for all identified risks
- ◆ A realistic 10-week project schedule has been defined and is achievable

Approval Signatures

| Role                        | Details                                          |
|-----------------------------|--------------------------------------------------|
| <i>Prepared By</i>          | Rabia Khalid   BSSE012415002                     |
| <i>Project Supervisor</i>   | Ma'am Aneela Abbas   University of Chenab        |
| <i>Institution</i>          | Department of CS, University of Chenab, Gujrat   |
| <i>Approval Status</i>      | APPROVED — Project Selected for Full Development |
| <i>Supervisor Signature</i> | _____                                            |
| <i>Date of Approval</i>     | _____                                            |

Step 02: Identify Project Scope & Objectives

This section defines the boundaries of the 3DPanoSphere project what is included, what is excluded, the key deliverables, system boundaries, and project assumptions.

Inclusions

The following features and modules are part of the 3DPanoSphere project and will be fully developed and delivered:

| #  | Feature                                        | Description                                                                                                    |
|----|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1  | <b><i>User Account Management</i></b>          | Registration, login (email/phone/social), profile management, 2FA, password reset, and logout from all devices |
| 2  | <b><i>Panorama Upload Module</i></b>           | Upload 360° panoramic images and videos from mobile device with compression options                            |
| 3  | <b><i>Panorama Management Module</i></b>       | Organize, edit, rename, categorize, delete, and apply filters to panoramic content                             |
| 4  | <b><i>Interactive 360° Viewer</i></b>          | Smooth viewer with touch gestures: swipe, zoom, rotate, and full-screen mode                                   |
| 5  | <b><i>Virtual Tours &amp; Storytelling</i></b> | Link multiple panoramas into sequential tours; add scene descriptions and navigation                           |
| 6  | <b><i>Interactive Hotspots Module</i></b>      | Add clickable hotspots inside panoramas with text, audio, or video popups                                      |
| 7  | <b><i>Social Interaction Module</i></b>        | Likes, comments, sharing, follow/unfollow, real-time messaging, group chats, block users, star ratings         |
| 8  | <b><i>Privacy &amp; Access Control</i></b>     | Public, private, or restricted visibility; content access permissions; account privacy settings                |
| 9  | <b><i>Search &amp; Discovery Module</i></b>    | Search by keywords, categories, hashtags; browse trending content; basic recommendation engine                 |
| 10 | <b><i>Profile Insights Module</i></b>          | View likes, comments, views, follower growth, and content popularity analytics                                 |
| 11 | <b><i>Notification Module</i></b>              | Push notifications for likes, comments, follows, messages, and platform announcements                          |
| 12 | <b><i>Admin &amp; Moderation Panel</i></b>     | Review reported content, manage user accounts, issue warnings or blocks, broadcast announcements               |
| 13 | <b><i>UI/UX Design</i></b>                     | Intuitive mobile design, onboarding tour, smooth navigation, accessible layouts                                |
| 14 | <b><i>Testing &amp; QA Reports</i></b>         | Unit testing, integration testing, UAT results, bug reports, and performance metrics                           |
| 15 | <b><i>Deployment Package</i></b>               | Final APK (Android) and IPA (iOS) ready for app store submission                                               |
| 16 | <b><i>Complete Documentation</i></b>           | SRS, user guide, technical documentation, and API documentation                                                |

## Exclusions

The following items are explicitly excluded from the current version of 3DPanoSphere. They may be considered for future releases:

| Excluded Item                                             | Reason                                                          |
|-----------------------------------------------------------|-----------------------------------------------------------------|
| <i>Web-based version of the platform</i>                  | Current focus is solely on mobile application (Android and iOS) |
| <i>Advanced AI image processing (auto-stitching, OCR)</i> | Technically complex; deferred to version 2.0                    |
| <i>VR headset or professional 360° camera integration</i> | Requires specialized hardware beyond current project scope      |
| <i>Payment gateway or subscription systems</i>            | No monetization features in the academic version                |
| <i>Enterprise-level cloud scaling / distributed infra</i> | Basic backend support is sufficient for academic project scale  |
| <i>Legal / trademark registration services</i>            | Not required for academic submission                            |
| <i>Automated content copyright detection</i>              | Advanced AI feature; deferred to future version                 |

### Identify Deliverables

The following are the formal deliverables that will be produced and submitted upon project completion:

| #  | Deliverable                                   | Completion Phase  | Format                |
|----|-----------------------------------------------|-------------------|-----------------------|
| 1  | <i>Fully Functional Mobile Application</i>    | Deployment Phase  | APK / IPA             |
| 2  | <i>Software Requirements Specification</i>    | Design Phase      | .docx / PDF           |
| 3  | <i>Business Case Document</i>                 | Step 1            | .docx / PDF           |
| 4  | <i>Project Charter</i>                        | Step 1            | .docx / PDF           |
| 5  | <i>Statement of Work (SOW)</i>                | Step 1            | .docx / PDF           |
| 6  | <i>UI/UX Mockups &amp; Prototypes</i>         | Design Phase      | Figma / PDF           |
| 7  | <i>Database Schema &amp; Architecture Doc</i> | Design Phase      | .docx / PDF           |
| 8  | <i>All 16 System Modules (Source Code)</i>    | Development Phase | Flutter + Spring Boot |
| 9  | <i>QA Testing Reports</i>                     | Testing Phase     | .docx / PDF           |
| 10 | <i>User Guide &amp; Training Material</i>     | Deployment Phase  | .docx / PDF           |
| 11 | <i>Final Presentation</i>                     | Project Close     | .pptx                 |

## Set System Boundaries

The system boundary defines what is inside the 3DPanoSphere system and what external systems it interacts with.

| Component                  | Inside System                                                     | Outside System / External                  |
|----------------------------|-------------------------------------------------------------------|--------------------------------------------|
| <i>User Interface</i>      | Flutter-based mobile app (Android & iOS)                          | Device operating system (Android / iOS)    |
| <i>Backend</i>             | Spring Boot REST API, authentication, business logic              | Internet service provider                  |
| <i>Database</i>            | MS SQL Server (user data, panoramic metadata, social data)        | Third-party analytics services             |
| <i>Media Storage</i>       | Cloud storage (AWS S3 / Firebase) for panoramic images and videos | CDN delivery network                       |
| <i>Notifications</i>       | Firebase Cloud Messaging (push notifications)                     | Email notification services                |
| <i>Authentication</i>      | JWT-based login, 2FA via OTP                                      | Google/Apple OAuth providers (optional)    |
| <i>Admin Panel</i>         | Web-accessible moderation interface                               | External reporting tools                   |
| <i>Testing Environment</i> | University lab devices, developer machines                        | Production app stores (Google Play, Apple) |

## Document Assumptions

The following assumptions have been made during the scoping of the 3DPanoSphere project:

| # | Assumption                                                                                | Impact if Not True                                               |
|---|-------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| 1 | Sufficient funding (\$2,680) will be available for infrastructure and deployment costs    | Development or deployment may be delayed or reduced in scope     |
| 2 | Users will have access to internet connectivity and compatible mobile devices             | App performance and usability will be impacted for offline users |
| 3 | Required tools (Flutter, Spring Boot, MS SQL Server) will remain available and free       | Additional licensing or migration costs may be incurred          |
| 4 | The developer (Rabia Khalid) will be available full-time throughout the project lifecycle | Project schedule may need to be revised                          |
| 5 | End users and businesses will be willing to adopt the platform                            | User adoption and platform success may be lower than projected   |
| 6 | The project will be completed within the 10-week academic timeline                        | Some advanced features may need to be deferred to v2.0           |

|   |                                                                                        |                                                      |
|---|----------------------------------------------------------------------------------------|------------------------------------------------------|
| 7 | University lab resources (devices, internet, lab space) will remain accessible         | Testing coverage on different devices may be limited |
| 8 | App store compliance requirements will be met for both Google Play and Apple App Store | Deployment to one or both platforms may be delayed   |

## Objectives

This section defines the project objectives in SMART format, the expected business benefits, success criteria, stakeholder expectations, and acceptance criteria.

### Measurable Goals (SMART Objectives)

Each objective is Specific, Measurable, Achievable, Relevant, and Time-bound:

| # | Objective                                           | Specific                                             | Measurable                                            | Target                     |
|---|-----------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|----------------------------|
| 1 | <b><i>Deliver a fully functional mobile app</i></b> | Develop all 16 modules for Android and iOS           | All modules functional and tested                     | By June 7, 2026            |
| 2 | <b><i>Achieve high system usability</i></b>         | Users can complete core tasks without assistance     | $\geq 85\%$ task completion rate in usability testing | Testing phase (Week 9)     |
| 3 | <b><i>Achieve strong user satisfaction</i></b>      | Collect in-app feedback from test users              | Average rating $\geq 4$ out of 5                      | After UAT                  |
| 4 | <b><i>Deliver high system performance</i></b>       | API and media loading must be fast and reliable      | Response time $< 3$ sec; media load $< 5$ sec         | Performance testing Week 9 |
| 5 | <b><i>Maintain low error rate</i></b>               | All modules should operate without critical failures | Error rate $< 3\%$ across all operations              | QA phase completion        |
| 6 | <b><i>Support concurrent users</i></b>              | App must handle multiple users at the same time      | Support $\geq 500$ concurrent users in load testing   | Load test Week 9           |
| 7 | <b><i>Enable virtual tour creation</i></b>          | Users can create and share multi-scene virtual tours | 1,000+ tours created within 3 months of deployment    | 3 months post-launch       |
| 8 | <b><i>Deploy to app stores</i></b>                  | App submitted to Google Play and Apple App Store     | Successful app store approval and installation        | Week 10                    |

### Business Benefits Expected

The following business benefits are expected from the successful delivery of 3DPanoSphere:

| Benefit Category                  | Expected Benefit                                                                        | Stakeholder               | Estimated Value |
|-----------------------------------|-----------------------------------------------------------------------------------------|---------------------------|-----------------|
| <b><i>Cost Savings</i></b>        | Eliminates the need for expensive physical visits and existing platform licensing fees  | Businesses, General Users | \$71,25         |
| <b><i>Revenue Potential</i></b>   | Enables future monetization through premium features, advertising, and B2B partnerships | Platform Owner            | \$92,50         |
| <b><i>User Productivity</i></b>   | Faster content creation and sharing compared to traditional methods                     | Content Creators          | \$17,50         |
| <b><i>Data Compliance</i></b>     | Built-in privacy controls and content moderation reduce legal risk                      | Platform Owner, Users     | \$12,00         |
| <b><i>Better Decisions</i></b>    | Analytics dashboard helps creators understand their audience and improve content        | Content Creators          | \$18,50         |
| <b><i>Reduced Maintenance</i></b> | Modular architecture lowers the long-term cost of system updates and fixes              | Developer, Platform Owner | \$26,00         |
| <b><i>TOTAL</i></b>               |                                                                                         |                           | \$23,775        |

### Success Criteria Defined

The project will be considered successful when all of the following criteria are met:

| #  | Success Criterion                                                                   | How It Will Be Measured              |
|----|-------------------------------------------------------------------------------------|--------------------------------------|
| 1  | <b><i>All 16 deliverables are completed and submitted</i></b>                       | Checklist review by supervisor       |
| 2  | <b><i>Mobile app works on both Android and iOS without critical bugs</i></b>        | Device compatibility and QA testing  |
| 3  | <b><i>System usability score <math>\geq 85\%</math></i></b>                         | Usability test with 100 participants |
| 4  | <b><i>User satisfaction rating <math>\geq 4/5</math></i></b>                        | In-app feedback survey post-testing  |
| 5  | <b><i>All API response times <math>&lt; 3</math> seconds under normal load</i></b>  | Performance testing tools            |
| 6  | <b><i>Error rate below 3% across all modules</i></b>                                | System logs and QA reports           |
| 7  | <b><i>360° panoramic viewer works smoothly on mid-range devices</i></b>             | Device-specific performance testing  |
| 8  | <b><i>Social features (messaging, likes, comments) are fully functional</i></b>     | Integration testing                  |
| 9  | <b><i>Admin moderation panel can approve/reject content and manage users</i></b>    | Manual acceptance testing            |
| 10 | <b><i>Project delivered within budget (\$2,680) and timeline (June 7, 2026)</i></b> | Budget tracking and milestone review |

## Stakeholder Expectations

Each key stakeholder group has specific expectations from the 3DPanoSphere project:

| Stakeholder                            | Role                             | Primary Expectation                                                                                         |
|----------------------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b><i>Rabia Khalid</i></b>             | Project Manager & Lead Developer | Deliver a fully working mobile app within the academic deadline and budget; earn strong project grade       |
| <b><i>Ma'am Aneela Abbas</i></b>       | Project Supervisor               | Regular milestone updates, professional documentation, and a well-tested final product                      |
| <b><i>University of Chenab</i></b>     | Academic Institution             | A completed project that demonstrates software engineering competency and follows BSSE curriculum standards |
| <b><i>Real Estate Businesses</i></b>   | End User / Client                | Easily create and share virtual property tours that reduce client visit costs                               |
| <b><i>Tourism Companies</i></b>        | End User / Client                | Showcase destinations with immersive 360° previews to attract online visitors                               |
| <b><i>Educational Institutions</i></b> | End User / Client                | Provide virtual campus tours and remote learning experiences for prospective students                       |
| <b><i>Content Creators</i></b>         | End User                         | Simple upload tools, audience analytics, and community features to grow their following                     |
| <b><i>General Users</i></b>            | End User                         | Easy-to-use app to explore and interact with immersive panoramic content                                    |
| <b><i>Peer Review Team</i></b>         | QA & Testing                     | Clear test plans, bug reports, and working modules ready for acceptance testing                             |

## Stakeholders

This section identifies all project stakeholders, assigns roles and authority, documents their interests, defines the communication plan, and outlines sign-off requirements.

### Identify All Stakeholders

| Stakeholder                                   | Type     | Interest Level | Influence Level |
|-----------------------------------------------|----------|----------------|-----------------|
| <b><i>Rabia Khalid (Project Manager)</i></b>  | Internal | Very High      | Very High       |
| <b><i>Ma'am Aneela Abbas (Supervisor)</i></b> | Internal | High           | High            |
| <b><i>University of Chenab (CS Dept.)</i></b> | Internal | Medium         | High            |
| <b><i>Peer Review / QA Team</i></b>           | Internal | Medium         | Medium          |

|                                                  |          |           |        |
|--------------------------------------------------|----------|-----------|--------|
| <b><i>Real Estate Businesses</i></b>             | External | High      | Medium |
| <b><i>Tourism Companies</i></b>                  | External | High      | Medium |
| <b><i>Educational Institutions</i></b>           | External | High      | Medium |
| <b><i>Content Creators / Photographers</i></b>   | External | Very High | Low    |
| <b><i>General End Users</i></b>                  | External | High      | Low    |
| <b><i>App Store Reviewers (Google/Apple)</i></b> | External | Low       | High   |

### Assign Roles & Authority

| Name / Party                       | Role                             | Authority Level | Responsibilities                                                                       |
|------------------------------------|----------------------------------|-----------------|----------------------------------------------------------------------------------------|
| <b><i>Rabia Khalid</i></b>         | Project Manager & Lead Developer | Full            | Plan, design, develop, test, and deploy entire application; manage schedule and budget |
| <b><i>Ma'am Aneela Abbas</i></b>   | Project Supervisor               | Approver        | Review milestones, approve deliverables, provide guidance and final sign-off           |
| <b><i>University of Chenab</i></b> | Academic Oversight               | Evaluator       | Evaluate project against BSSE curriculum standards; approve final submission           |
| <b><i>Peer Review Team</i></b>     | QA & Testing Support             | Reviewer        | Test modules, report bugs, provide usability feedback                                  |
| <b><i>End Users (Beta)</i></b>     | Beta Testers                     | Feedback        | Test app features and provide usability and satisfaction feedback                      |
| <b><i>App Stores</i></b>           | Distribution Platform            | Gatekeeper      | Review and approve app for public distribution on Google Play and Apple App Store      |

### Document Stakeholder Interests

| Stakeholder                        | Primary Interest             | Secondary Interest                            | Concern / Risk                                |
|------------------------------------|------------------------------|-----------------------------------------------|-----------------------------------------------|
| <b><i>Rabia Khalid</i></b>         | Successful academic grade    | Building a portfolio-worthy project           | Time pressure as solo developer               |
| <b><i>Ma'am Aneela Abbas</i></b>   | High-quality academic output | Student learning and professional development | Missed milestones or incomplete documentation |
| <b><i>University of Chenab</i></b> | Academic standard compliance | Reputation in software engineering education  | Project not meeting curriculum requirements   |



|                                        |                                   |                                            |                                         |
|----------------------------------------|-----------------------------------|--------------------------------------------|-----------------------------------------|
| <b><i>Real Estate Businesses</i></b>   | Virtual property showcasing       | Reducing costs of physical property visits | Platform adoption and reliability       |
| <b><i>Tourism Companies</i></b>        | Destination promotion             | Reaching international audiences online    | Content quality and viewer experience   |
| <b><i>Educational Institutions</i></b> | Remote campus tours               | Engaging prospective students digitally    | Privacy of student data                 |
| <b><i>Content Creators</i></b>         | Community and audience growth     | Monetization potential in future           | Platform stability and feature richness |
| <b><i>General Users</i></b>            | Immersive and free content access | Social interaction and discovery           | Privacy and ease of use                 |

### Communication Plan

| Stakeholder                       | Communication Type                      | Frequency       | Channel                         | Responsible  |
|-----------------------------------|-----------------------------------------|-----------------|---------------------------------|--------------|
| <b><i>Ma'am Aneela Abbas</i></b>  | Progress update & milestone review      | Weekly          | In-person meeting / Email       | Rabia Khalid |
| <b><i>University CS Dept.</i></b> | Formal submission of deliverables       | Per milestone   | Printed & digital submission    | Rabia Khalid |
| <b><i>Peer Review Team</i></b>    | Bug reports & testing feedback          | During testing  | WhatsApp / Slack / Lab sessions | Rabia Khalid |
| <b><i>Beta Users</i></b>          | Usability testing & feedback survey     | Once (Week 9)   | In-app form / Google Form       | Rabia Khalid |
| <b><i>Supervisor</i></b>          | Final presentation                      | Once (Week 10)  | University presentation room    | Rabia Khalid |
| <b><i>All Stakeholders</i></b>    | Project retrospective & lessons learned | Post-completion | Formal report                   | Rabia Khalid |

### Sign-Off Requirements

The following sign-offs are required before the project is formally closed:

| Deliverable / Milestone                              | Sign-Off Required From        | Method                        | Status  |
|------------------------------------------------------|-------------------------------|-------------------------------|---------|
| <b><i>Step 2 Scope &amp; Objectives Document</i></b> | Ma'am Aneela Abbas            | Signature on document         | Pending |
| <b><i>System Architecture &amp; DB Design</i></b>    | Ma'am Aneela Abbas            | Email approval                | Pending |
| <b><i>UI/UX Mockups</i></b>                          | Ma'am Aneela Abbas            | Review meeting                | Pending |
| <b><i>Core Modules Development</i></b>               | Peer Review Team + Supervisor | Acceptance test sign-off      | Pending |
| <b><i>QA &amp; Testing Reports</i></b>               | Ma'am Aneela Abbas            | Document review and signature | Pending |

|                                        |                                 |                             |         |
|----------------------------------------|---------------------------------|-----------------------------|---------|
| <b><i>Final Mobile Application</i></b> | Ma'am Aneela Abbas + University | Final presentation sign-off | Pending |
| <b><i>Project Documentation</i></b>    | Ma'am Aneela Abbas              | Formal submission receipt   | Pending |

### Step 3: Identify Project Infrastructure

#### Dependencies and Shared Resources

3DPanoSphere is a standalone academic project. However, it relies on the following internal documents and external technologies throughout its lifecycle.

#### Internal Document Dependencies

| Document                                                | Purpose                                                      |
|---------------------------------------------------------|--------------------------------------------------------------|
| <b><i>Software Requirements Specification (SRS)</i></b> | Defines all functional and non-functional requirements       |
| <b><i>Business Case</i></b>                             | Justifies the investment and sets expected benefits          |
| <b><i>Project Charter</i></b>                           | Formally authorizes the project; defines boundaries and KPIs |
| <b><i>Statement of Work (SOW)</i></b>                   | Ties deliverables to milestones and payment schedule         |

#### External Technology Dependencies

| Technology / Service                          | Role                                    | Risk if Changed                                   |
|-----------------------------------------------|-----------------------------------------|---------------------------------------------------|
| <b><i>Flutter SDK (Google)</i></b>            | Mobile frontend development             | Major version update may require code changes     |
| <b><i>Spring Boot (Java)</i></b>              | Backend REST API development            | Framework version incompatibility with SQL Server |
| <b><i>MS SQL Server Developer Edition</i></b> | Primary relational database             | Migration would be costly if replaced             |
| <b><i>AWS S3 / Firebase Storage</i></b>       | Cloud storage for panoramic media files | Pricing or API change affects media pipeline      |
| <b><i>Firebase Cloud Messaging (FCM)</i></b>  | Push notification delivery              | Service outage disrupts all notifications         |
| <b><i>Google Play / Apple App Store</i></b>   | App distribution platforms              | Policy changes can delay deployment               |
| <b><i>GitHub</i></b>                          | Version control and code backup         | Shared with supervisor for milestone review       |

Standards and Procedures

| Standard                                  | Category        | Application                                            |
|-------------------------------------------|-----------------|--------------------------------------------------------|
| <i>IEEE 830</i>                           | Documentation   | SRS structure and completeness                         |
| <i>Agile (2-week sprints)</i>             | Dev Process     | Iterative builds with supervisor review each sprint    |
| <i>Git / GitHub Branching Strategy</i>    | Version Control | Feature branches per module; main branch = stable code |
| <i>REST API Design Principles</i>         | Development     | HTTP methods, status codes, JSON responses             |
| <i>Material Design 3 (Google)</i>         | UI/UX           | Accessible, consistent mobile interface                |
| <i>OWASP Mobile Top-10</i>                | Security        | JWT auth, input validation, encrypted storage, HTTPS   |
| <i>GDPR Best Practices</i>                | Legal           | User consent, secure storage, right to data deletion   |
| <i>Google Play / Apple Store Policies</i> | Deployment      | Content, privacy, and performance compliance           |

Installations — Hardware, Software, and Network

Hardware

| Item                                    | Purpose                                      | Availability              |
|-----------------------------------------|----------------------------------------------|---------------------------|
| <i>Developer Laptop / PC</i>            | All coding, testing, and documentation       | Personal device           |
| <i>Android Smartphones (2–3 models)</i> | Cross-device compatibility testing           | University lab + personal |
| <i>iPhone</i>                           | iOS platform testing                         | University lab (shared)   |
| <i>360° Camera (optional)</i>           | Capturing real panoramic content for testing | Can substitute smartphone |

Software and Development Tools

| Tool                            | Purpose                         | Cost               |
|---------------------------------|---------------------------------|--------------------|
| <i>Flutter SDK</i>              | Mobile frontend development     | Free (open-source) |
| <i>Spring Boot (Java)</i>       | Backend REST API                | Free (open-source) |
| <i>MS SQL Server Dev. Ed.</i>   | Primary database                | Free               |
| <i>Android Studio / VS Code</i> | IDE for Flutter and Java        | Free               |
| <i>Figma (free tier)</i>        | UI/UX wireframes and prototypes | Free               |
| <i>Postman</i>                  | REST API testing                | Free               |

| Tool                | Purpose                    | Cost |
|---------------------|----------------------------|------|
| <i>Git / GitHub</i> | Version control and backup | Free |

Cloud and Network Infrastructure

| Item                        | Specification                                    | Cost        |
|-----------------------------|--------------------------------------------------|-------------|
| <i>Cloud Backend Server</i> | Basic VPS (1 vCPU, 1GB RAM) — AWS / DigitalOcean | ~\$20/month |
| <i>Cloud Media Storage</i>  | 500 GB — AWS S3 / Firebase Storage               | ~\$30/month |
| <i>Domain Name</i>          | .com domain for backend API                      | \$15/year   |
| <i>SSL Certificate</i>      | HTTPS via Let's Encrypt                          | Free        |
| <i>University Wi-Fi</i>     | Development and testing internet                 | Provided    |
| <i>GitHub Actions</i>       | CI/CD automation (optional)                      | Free        |

Team Structure — Who Reports to Whom

| Name / Party                    | Role                             | Reports To                      | Responsibilities                                                                            |
|---------------------------------|----------------------------------|---------------------------------|---------------------------------------------------------------------------------------------|
| <i>Ma'am Aneela Abbas</i>       | Project Supervisor               | University of Chenab (CS Dept.) | Review milestones, approve deliverables, provide guidance, give final sign-off              |
| <i>Rabia Khalid</i>             | Project Manager & Lead Developer | Ma'am Aneela Abbas              | Plan, design, develop, test, and deploy all modules; manage schedule, budget, documentation |
| <i>Peer Review Team</i>         | QA & Testing Support             | Rabia Khalid                    | Perform testing, report bugs, give usability feedback on modules                            |
| <i>Beta Testers (End Users)</i> | Usability Testers                | Rabia Khalid                    | Test app features, complete usability surveys, report issues found                          |
| <i>University CS Dept.</i>      | Academic Oversight               | University Administration       | Evaluate and grade the final project; ensure BSSE curriculum compliance                     |

**Reporting Flow:** *Beta Testers → Rabia Khalid → Ma'am Aneela Abbas → University of Chenab (CS Dept.)*

Step 4: Analyse Project Characteristics

Well-Defined or Exploratory?

3DPanoSphere is primarily a well-defined project. Scope, deliverables, and technologies are clearly documented in the SRS, SOW, and Business Case. A small number of exploratory elements exist — primarily 360° viewer performance on low-end devices and real-time messaging optimization. These will be resolved through early prototyping in Sprint 1 (Week 4) before full development begins.

Application Type

| Type                      | Applies?      | How It Applies to 3DPanoSphere                                                                                                  |
|---------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| <i>Data-Intensive</i>     | YES — High    | Panoramic files are 10–100MB+. The system handles bulk uploads, media streaming, user profiles, social data, and analytics.     |
| <i>Real-Time</i>          | YES — Medium  | Live messaging (WebSocket/Firebase), push notifications (FCM), and live content updates require event-driven backend design.    |
| <i>Highly Interactive</i> | YES — High    | 360° viewer uses swipe, zoom, rotate gestures. Hotspots, virtual tours, and social features all demand a smooth, responsive UI. |
| <i>Mobile-First</i>       | YES — Primary | Built with Flutter for Android (API 21+) and iOS (14+). No web version in current scope.                                        |
| <i>Cloud-Based</i>        | YES — High    | Backend, media storage, and push notifications are all cloud-hosted. Internet connectivity is required.                         |
| <i>AI / ML</i>            | Partial       | Basic recommendations included in Search & Discovery. Advanced AI (auto-stitching, object recognition) deferred to v2.0.        |
| <i>Embedded</i>           | No            | No IoT or embedded hardware. Only standard smartphone sensors (gyroscope) are used.                                             |

Build or Buy Decision

The guiding principle: build what requires full control; use managed services where building from scratch adds no value.

| Component                   | Decision                     | Justification                                                   |
|-----------------------------|------------------------------|-----------------------------------------------------------------|
| <i>Mobile App Frontend</i>  | Build — Flutter              | Free, cross-platform. Full control over UI and 360° viewer.     |
| <i>Backend REST API</i>     | Build — Spring Boot          | Full control over business logic, authentication, and database. |
| <i>360° Panorama Viewer</i> | Build with library support   | Custom viewer needed to support hotspot overlays.               |
| <i>Database</i>             | Use — MS SQL Server Dev. Ed. | Free and fully functional. No need to build a custom DB engine. |

| Component                         | Decision                     | Justification                                                                        |
|-----------------------------------|------------------------------|--------------------------------------------------------------------------------------|
| <i>Cloud Storage</i>              | Use — AWS S3 / Firebase      | Managed service. Building cloud storage infrastructure is not feasible academically. |
| <i>Push Notifications</i>         | Use — Firebase FCM           | Free, reliable, cross-platform. Building from scratch is impractical.                |
| <i>Authentication (JWT + 2FA)</i> | Build — Java libraries       | Standard libraries are sufficient. No commercial auth service needed.                |
| <i>Real-Time Messaging</i>        | Build — WebSocket / Firebase | Avoids expensive third-party chat SDK licensing.                                     |
| <i>Admin Panel</i>                | Build                        | Custom panel needed for full control over moderation and user management.            |
| <i>CI/CD Pipeline</i>             | Use — GitHub Actions         | Free tier handles automated builds and tests. No custom setup needed.                |

Development Methodology — Agile

Agile methodology with 2-week sprints was selected because the project has mostly stable requirements but a few technical uncertainties that benefit from short feedback cycles. As a solo developer with a fixed academic deadline, a lightweight process that avoids heavy overhead is essential. Agile also supports incremental delivery — core modules are built first, advanced features come in later sprints, so there is always a working version to demonstrate.

| Sprint          | Weeks | Focus                      | Key Deliverables                                                         |
|-----------------|-------|----------------------------|--------------------------------------------------------------------------|
| <i>Sprint 1</i> | 1–2   | Planning & Design          | Requirements, architecture, DB schema, UI/UX mockups                     |
| <i>Sprint 2</i> | 3–4   | Core Setup & User Module   | Dev environment, user auth, login and registration screens               |
| <i>Sprint 3</i> | 5–6   | Panorama & Viewer Modules  | Upload module, panorama management, interactive 360° viewer              |
| <i>Sprint 4</i> | 7–8   | Social & Advanced Features | Virtual tours, hotspots, social features, search, notifications, admin   |
| <i>Sprint 5</i> | 9–10  | Testing & Deployment       | Integration, QA testing, bug fixing, app store deployment, documentation |

Waterfall was considered but rejected — it requires all requirements finalized before development begins and has no mechanism for mid-project feedback. Agile handles the remaining technical uncertainties far better by building learning and adjustment into each sprint.

Step 5: Identify Products & Activities (Work Breakdown)

Hierarchical WBS

The Hierarchical WBS follows the IBM MITP five-level framework. The project is decomposed from a single top-level entity down to individual tasks. Each level adds more detail and granularity. This structure is the core reference for all planning, estimation, and tracking activities.

| ID    | Level 2: Deliverable   | Level 3: Component                 | Level 4: Work Package                          |
|-------|------------------------|------------------------------------|------------------------------------------------|
| 2.1   | User Management System |                                    |                                                |
| 2.1.1 |                        | User Registration & Authentication |                                                |
|       |                        |                                    | 2.1.1.1 User Registration (Email/Phone/Social) |
|       |                        |                                    | 2.1.1.2 User Login (with 2FA)                  |
|       |                        |                                    | 2.1.1.3 Password Reset                         |
|       |                        |                                    | 2.1.1.4 Session Management                     |
| 2.1.2 |                        | User Profile Management            |                                                |
|       |                        |                                    | 2.1.2.1 Profile Creation                       |
|       |                        |                                    | 2.1.2.2 Profile Picture Upload                 |
|       |                        |                                    | 2.1.2.3 Profile Editing & Updates              |
|       |                        |                                    | 2.1.2.4 Account Deactivation / Deletion        |
| 2.1.3 |                        | Account Security & Privacy         |                                                |
|       |                        |                                    | 2.1.3.1 Account Privacy Controls               |
|       |                        |                                    | 2.1.3.2 Content Visibility Settings            |
|       |                        |                                    | 2.1.3.3 Data Sharing Preferences               |
|       |                        |                                    | 2.1.3.4 Account Activity Monitoring            |
| 2.1.4 |                        | Guest User Access                  |                                                |
|       |                        |                                    | 2.1.4.1 Browse Public Content                  |
|       |                        |                                    | 2.1.4.2 Limited Story Viewing                  |
|       |                        |                                    | 2.1.4.3 Search Preview                         |
|       |                        |                                    | 2.1.4.4 App Tour / Onboarding                  |

| ID           | Level 2: Deliverable             | Level 3: Component                       | Level 4: Work Package                       |
|--------------|----------------------------------|------------------------------------------|---------------------------------------------|
| <b>2.2</b>   | <b>Panoramic Content System</b>  |                                          |                                             |
| <b>2.2.1</b> |                                  | <b>Panorama Capture &amp; Creation</b>   |                                             |
|              |                                  |                                          | 2.2.1.1 Capture Multiple Overlapping Images |
|              |                                  |                                          | 2.2.1.2 Auto-Stitching Engine               |
|              |                                  |                                          | 2.2.1.3 Panorama Preview & Save             |
| <b>2.2.2</b> |                                  | <b>Panorama Viewing &amp; Navigation</b> |                                             |
|              |                                  |                                          | 2.2.2.1 Full-screen Interactive Viewer      |
|              |                                  |                                          | 2.2.2.2 Zoom, Rotate & Swipe Navigation     |
|              |                                  |                                          | 2.2.2.3 Star Rating for Panoramas           |
| <b>2.2.3</b> |                                  | <b>Media Upload &amp; Management</b>     |                                             |
|              |                                  |                                          | 2.2.3.1 Upload Image/Video                  |
|              |                                  |                                          | 2.2.3.2 Add Captions & Tags                 |
|              |                                  |                                          | 2.2.3.3 Post Visibility Settings            |
|              |                                  |                                          | 2.2.3.4 Delete Posts                        |
| <b>2.2.4</b> |                                  | <b>Content Editing &amp; Filters</b>     |                                             |
|              |                                  |                                          | 2.2.4.1 Apply Filters & Effects             |
|              |                                  |                                          | 2.2.4.2 Panorama Sharing                    |
| <b>2.3</b>   | <b>Social Interaction System</b> |                                          |                                             |
| <b>2.3.1</b> |                                  | <b>Likes &amp; Comments</b>              |                                             |
|              |                                  |                                          | 2.3.1.1 Like/Unlike Panoramas               |
|              |                                  |                                          | 2.3.1.2 Comment on Posts                    |
|              |                                  |                                          | 2.3.1.3 Share Panoramas                     |
| <b>2.3.2</b> |                                  | <b>Follow / Unfollow System</b>          |                                             |
|              |                                  |                                          | 2.3.2.1 Follow/Unfollow Users               |



| ID    | Level 2: Deliverable          | Level 3: Component         | Level 4: Work Package                      |
|-------|-------------------------------|----------------------------|--------------------------------------------|
|       |                               |                            | 2.3.2.2 Block/Unblock Users                |
| 2.3.3 |                               | Messaging System           |                                            |
|       |                               |                            | 2.3.3.1 Direct Messages (Text/Audio/Video) |
|       |                               |                            | 2.3.3.2 Group Creation & Management        |
|       |                               |                            | 2.3.3.3 Pin/Delete Chats                   |
| 2.3.4 |                               | Stories System             |                                            |
|       |                               |                            | 2.3.4.1 Create Time-limited Stories        |
|       |                               |                            | 2.3.4.2 Story Customization                |
|       |                               |                            | 2.3.4.3 Story Highlights                   |
|       |                               |                            | 2.3.4.4 Delete Story                       |
| 2.4   | Virtual Gallery & Tour System |                            |                                            |
| 2.4.1 |                               | Virtual Gallery Management |                                            |
|       |                               |                            | 2.4.1.1 Create Virtual Gallery             |
|       |                               |                            | 2.4.1.2 Customize Gallery Layouts & Themes |
|       |                               |                            | 2.4.1.3 Organize & Curate Content          |
|       |                               |                            | 2.4.1.4 Visitor Analytics                  |
| 2.4.2 |                               | Virtual Tour Creation      |                                            |
|       |                               |                            | 2.4.2.1 Link Multiple Panoramas            |
|       |                               |                            | 2.4.2.2 Add Scene Descriptions             |
|       |                               |                            | 2.4.2.3 Enable Navigation Between Scenes   |
| 2.4.3 |                               | Interactive Hotspots       |                                            |
|       |                               |                            | 2.4.3.1 Add Text Hotspots                  |
|       |                               |                            | 2.4.3.2 Add Image Hotspots                 |
|       |                               |                            | 2.4.3.3 Add Audio/Video Hotspots           |
| 2.5   | Analytics & Insights System   |                            |                                            |

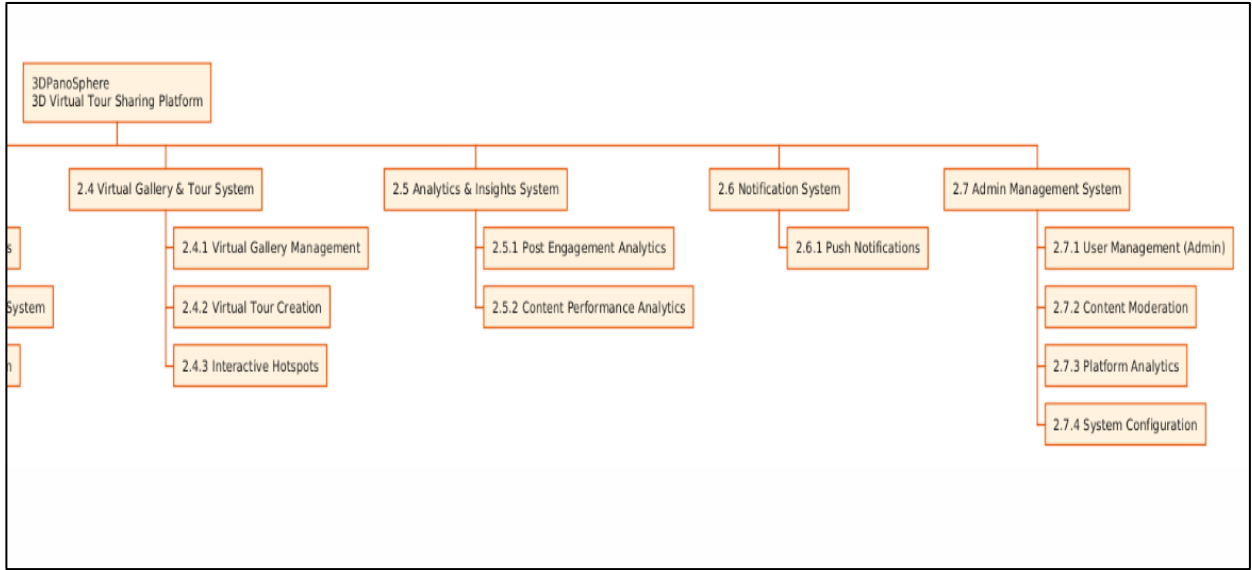
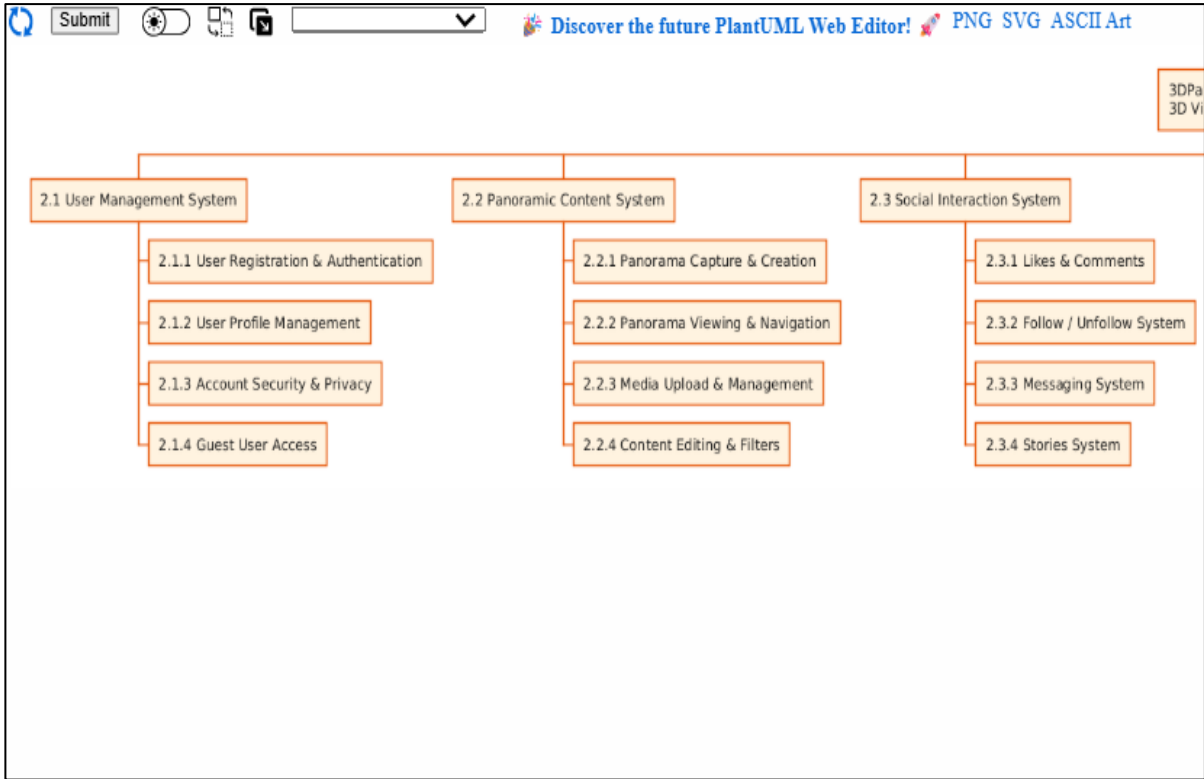
| ID    | Level 2: Deliverable    | Level 3: Component            | Level 4: Work Package                       |
|-------|-------------------------|-------------------------------|---------------------------------------------|
| 2.5.1 |                         | Post Engagement Analytics     |                                             |
|       |                         |                               | 2.5.1.1 View Post Engagement Metrics        |
|       |                         |                               | 2.5.1.2 Content Performance Analysis        |
|       |                         |                               | 2.5.1.3 Best Posting Time Recommendations   |
| 2.5.2 |                         | Content Performance Analytics |                                             |
|       |                         |                               | 2.5.2.1 Profile Visit Analytics             |
|       |                         |                               | 2.5.2.2 Content Reach Statistics            |
|       |                         |                               | 2.5.2.3 Audience Growth Tracking            |
| 2.6   | Notification System     |                               |                                             |
| 2.6.1 |                         | Push Notifications            |                                             |
|       |                         |                               | 2.6.1.1 Push Notifications for Interactions |
|       |                         |                               | 2.6.1.2 Custom Notification Preferences     |
|       |                         |                               | 2.6.1.3 Real-time Engagement Alerts         |
|       |                         |                               | 2.6.1.4 Important Updates & Announcements   |
| 2.7   | Admin Management System |                               |                                             |
| 2.7.1 |                         | User Management (Admin)       |                                             |
|       |                         |                               | 2.7.1.1 View All Registered Users           |
|       |                         |                               | 2.7.1.2 Block/Unblock Users                 |
|       |                         |                               | 2.7.1.3 Delete User Accounts                |
|       |                         |                               | 2.7.1.4 Verify User Accounts                |
| 2.7.2 |                         | Content Moderation            |                                             |
|       |                         |                               | 2.7.2.1 Review Reported Content             |
|       |                         |                               | 2.7.2.2 Warn or Block Violating Users       |
|       |                         |                               | 2.7.2.3 Manage Appeals                      |
|       |                         |                               | 2.7.2.4 Enforce Community Guidelines        |

| ID    | Level 2: Deliverable | Level 3: Component         | Level 4: Work Package                      |
|-------|----------------------|----------------------------|--------------------------------------------|
| 2.7.3 |                      | Platform Analytics (Admin) |                                            |
|       |                      |                            | 2.7.3.1 Platform Usage Analytics           |
|       |                      |                            | 2.7.3.2 Content Engagement Reports         |
|       |                      |                            | 2.7.3.3 User Activity Reports              |
| 2.7.4 |                      | System Configuration       |                                            |
|       |                      |                            | 2.7.4.1 Update System Configurations       |
|       |                      |                            | 2.7.4.2 Manage Content Templates & Filters |
|       |                      |                            | 2.7.4.3 System Backup and Restore          |
|       |                      |                            | 2.7.4.4 Broadcast Important Updates        |

Level 5 Example (Task Breakdown)

Example 1 — Work Package 2.1.1.1: User Registration (Email/Phone/Social)

| Task ID    | Task                                             |
|------------|--------------------------------------------------|
| 2.1.1.1.1  | Design Registration Form UI                      |
| 2.1.1.1.2  | Create Backend API for Registration              |
| 2.1.1.1.3  | Implement Email Verification                     |
| 2.1.1.1.4  | Implement Phone Verification                     |
| 2.1.1.1.5  | Integrate Social Media Sign-In (Google/Facebook) |
| 2.1.1.1.6  | Update Database (Users Table)                    |
| 2.1.1.1.7  | Implement Input Validation                       |
| 2.1.1.1.8  | Write Unit Tests                                 |
| 2.1.1.1.9  | Perform Integration Testing                      |
| 2.1.1.1.10 | Fix Bugs                                         |



Dependencies Between Activities

Understanding dependencies between activities is critical for building a realistic schedule. A dependency means that one activity cannot begin until another is complete. In 3DPanoSphere, these dependencies follow both technical logic (you cannot test what has not been built) and architectural logic (all modules depend on the user auth module being in place first).

| Dependency         | From → To                                   | Reason                                                                                        |
|--------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------|
| <i>D1 → D2</i>     | Requirements Document → System Design       | Design cannot begin until requirements are finalized and approved                             |
| <i>D2 → D3</i>     | System Design → Development (all 7 modules) | Development of each module depends on completed architecture and DB design                    |
| <i>D3.1 → D3.2</i> | User Management Module → All other modules  | All modules depend on authentication; users must exist before content or social features work |
| <i>D3.2 ↔ D3.4</i> | Panoramic Content ↔ Virtual Gallery & Tour  | Gallery and tour features depend on panoramas being uploadable and viewable                   |

|                         |                                        |                                                                                       |
|-------------------------|----------------------------------------|---------------------------------------------------------------------------------------|
| <b>D3.2 → D3.3</b>      | Panoramic Content → Social Interaction | Likes, comments, and shares only make sense once panoramic content exists             |
| <b>D3.3 → D3.5</b>      | Social Interaction → Analytics         | Engagement analytics require social actions (likes, follows, comments) to be in place |
| <b>D3.1–D3.7 → D4</b>   | All 7 modules → Integration & Testing  | Full system testing can only begin once all modules are individually developed        |
| <b>D4 → D5</b>          | Testing → Deployment                   | Deployment happens only after QA testing passes and bugs are resolved                 |
| <b>D3.6 → D3.1</b>      | Notification Module → User Management  | Push notifications require user accounts and FCM tokens registered at login           |
| <b>D3.7 → D3.1–D3.5</b> | Admin Panel → All user modules         | Admin moderation and analytics depend on all user-facing modules being functional     |

The most critical dependency chain in this project is: Requirements → Design → User Auth Module → All other modules → Integration → Testing → Deployment. This forms the longest critical path and any delay in this chain will delay the entire project.

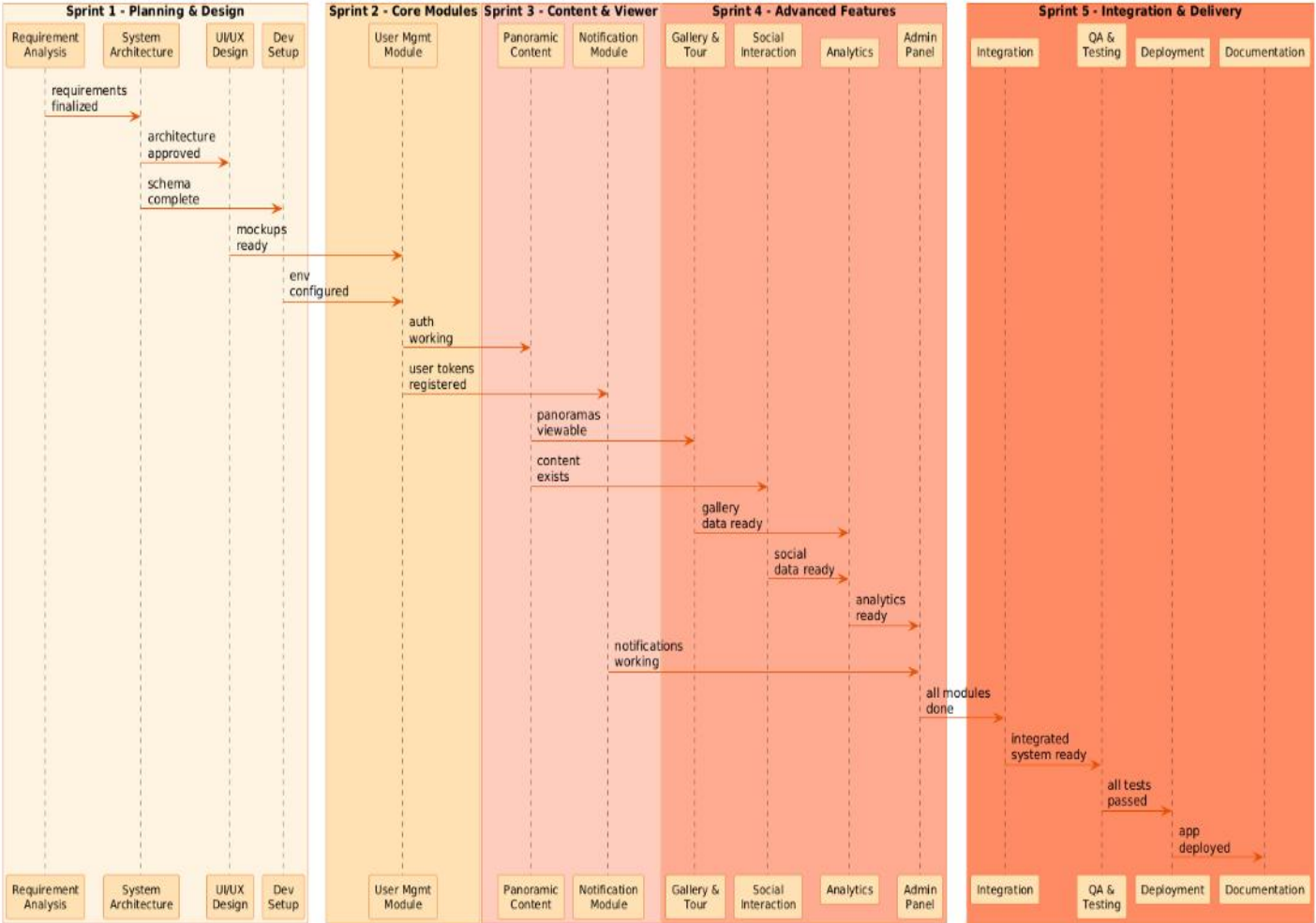
### Activity Sequencing & Precedence Analysis

Precedence analysis organizes all activities in the correct order, taking into account their dependencies. The table below shows each activity, what it depends on, when it can start, and what it produces. This forms the basis for the project schedule and Gantt chart.

| # | Activity                                               | Depends On     | Can Start When                | Deliverable                         |
|---|--------------------------------------------------------|----------------|-------------------------------|-------------------------------------|
| 1 | <b>Requirement Analysis &amp; Feature Finalization</b> | None           | Start                         | Requirements Document, Feature List |
| 2 | <b>System Architecture &amp; Database Design</b>       | Activity 1     | Activity 1 complete           | Architecture Doc, DB Schema         |
| 3 | <b>UI/UX Design &amp; Prototyping</b>                  | Activity 2     | Activity 2 complete           | Figma Mockups, Prototypes           |
| 4 | <b>Dev Environment Setup</b>                           | Activity 2     | Activity 2 complete           | Configured dev environment          |
| 5 | <b>User Management Module Development</b>              | Activities 3–4 | Activities 3 & 4 complete     | Working auth & profile module       |
| 6 | <b>Panoramic Content Module Development</b>            | Activity 5     | User module done              | Upload, viewer, editing working     |
| 7 | <b>Virtual Gallery &amp; Tour Module Development</b>   | Activity 6     | Panoramic module done         | Gallery & tour features working     |
| 8 | <b>Social Interaction Module Development</b>           | Activity 6     | Panoramic module done         | Likes, comments, messaging, stories |
| 9 | <b>Analytics &amp; Insights Module Development</b>     | Activities 7–8 | Social & gallery modules done | Analytics dashboard working         |

|    |                                         |                 |                                 |                               |
|----|-----------------------------------------|-----------------|---------------------------------|-------------------------------|
| 10 | Notification Module Development         | Activity 5      | User module done                | Push notifications working    |
| 11 | Admin Panel Module Development          | Activities 5–10 | All user modules done           | Full admin panel working      |
| 12 | Integration of All Modules              | Activities 5–11 | All 7 modules individually done | Fully integrated application  |
| 13 | QA Testing & Bug Fixing                 | Activity 12     | Integration complete            | Test reports, resolved bugs   |
| 14 | Final Deployment & App Store Submission | Activity 13     | All tests passed                | APK/IPA deployed              |
| 15 | Documentation & Project Close           | Activity 14     | Deployment complete             | Final docs, submitted project |

..11 ATT



## Step 6: Estimate Effort, Duration & Schedule

### Estimation Techniques Used

Estimating effort, duration, and cost accurately is one of the most important activities in software project management. If estimates are too optimistic, the project runs late and over budget. If they are too pessimistic, resources are wasted. For 3DPanoSphere, a combination of four estimation techniques was used to cross-validate the results and arrive at reliable figures.

| Technique                     | Approach                                                                         | How Applied to 3DPanoSphere                                                                                                                                              |
|-------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Expert Judgement</i>       | Ask experienced staff to estimate based on similar past work                     | The developer (Rabia Khalid) estimated each module based on her experience with Flutter, Spring Boot, and SQL Server from previous academic projects and labs            |
| <i>Analogy Based</i>          | Compare to a similar project completed previously                                | The project was compared to similar academic mobile app projects (social platforms, content-sharing apps) of similar scope and complexity to calibrate estimates         |
| <i>Parametric (COCOMO II)</i> | Use a mathematical model (COCOMO II formula) to calculate size, effort, and time | COCOMO II was applied in Lab 09 using 26 KLOC project size, EAF = 1.25, and B = 1.10 to calculate effort (45 person-months), duration (10.5 months), and cost (\$54,000) |
| <i>Bottom-Up Estimation</i>   | Estimate each task individually from the WBS, then sum the total                 | Each of the 15 activities from the WBS (Step 5) was estimated individually in person-days, then summed to get the total project effort                                   |

### Parametric Estimation — Cocomo II

COCOMO II (Constructive Cost Model II) is a formal parametric estimation model that calculates effort, development time, and team size based on the project size in KLOC (thousands of lines of code) and a set of cost and scale drivers. This was applied in Lab 09 specifically for 3DPanoSphere.

### Project Size & Formula

The 3DPanoSphere project was estimated at 26 KLOC (26,000 lines of source code). This accounts for all 7 modules: User Management, Panoramic Content, Social Interaction, Virtual Gallery & Tour, Analytics, Notifications, and Admin Panel, including frontend (Flutter), backend (Spring Boot), and database layer.

| Parameter                  | Value | Meaning                                                                                      |
|----------------------------|-------|----------------------------------------------------------------------------------------------|
| <i>Project Size (KLOC)</i> | 26    | 26,000 estimated lines of code across all 7 modules                                          |
| <i>A (constant)</i>        | 2.94  | COCOMO II baseline constant for effort calculation                                           |
| <i>B (scaling factor)</i>  | 1.10  | Reflects semi-detached project with medium complexity                                        |
| <i>EAF (Effort Adj.)</i>   | 1.25  | Product of all cost driver ratings — reflects high complexity, DB size, and time constraints |

### Effort Calculation

The COCOMO II effort formula is:  $\text{Effort} = A \times (\text{Size})^B \times \text{EAF}$

Substituting the values:  $\text{Effort} = 2.94 \times (26)^{1.10} \times 1.25$

**Estimated Effort = 45 Person-Months**

This means that if one person worked on this project full-time, it would take approximately 45 months to complete. With a team of 4–5 developers working in parallel, the calendar time reduces significantly.

### Development Time Calculation

The development time formula is:  $\text{Time} = 3.67 \times (\text{Effort})^{0.28}$

Substituting:  $\text{Time} = 3.67 \times (45)^{0.28}$

**Estimated Development Time = 10.5 Months**

### Team Size Calculation

Team Size = Effort ÷ Development Time =  $45 \div 10.5$

**Estimated Team Size = 4–5 Developers**

### COCOMO II Scale & Cost Drivers

| Driver                               | Value   | Category            |
|--------------------------------------|---------|---------------------|
| <i>Precedentedness</i>               | Nominal | Scale Driver        |
| <i>Development Flexibility</i>       | Nominal | Scale Driver        |
| <i>Architecture/Risk Resolution</i>  | Nominal | Scale Driver        |
| <i>Team Cohesion</i>                 | High    | Scale Driver        |
| <i>Process Maturity</i>              | Low     | Scale Driver        |
| <i>Required Software Reliability</i> | Nominal | Product Cost Driver |
| <i>Database Size</i>                 | High    | Product Cost Driver |
| <i>Product Complexity</i>            | High    | Product Cost Driver |
| <i>Time Constraint</i>               | High    | Platform Driver     |
| <i>Multisite Development</i>         | Low     | Project Driver      |
| <i>Required Development Schedule</i> | High    | Project Driver      |

### Bottom-Up Estimation (Per Activity)

Bottom-up estimation assigns effort (in person-days) to each individual activity from the WBS. The total is then summed to get the overall project effort. This is the most detailed and accurate method for a well-defined project like 3DPanoSphere.



All estimates assume a solo developer (Rabia Khalid) working approximately 6–8 hours per day in an academic setting. One person-day = 7 hours of productive work.

| #  | Activity / Deliverable                                 | Effort (Person-Days) | Duration (Calendar Days) | Notes                                                             |
|----|--------------------------------------------------------|----------------------|--------------------------|-------------------------------------------------------------------|
| 1  | <b>Requirement Analysis &amp; Feature Finalization</b> | 5                    | 5                        | Review SRS, finalize feature list, get supervisor approval        |
| 2  | <b>System Architecture &amp; Database Design</b>       | 4                    | 5                        | Design backend, DB schema, API contracts, cloud architecture      |
| 3  | <b>UI/UX Design &amp; Prototyping (Figma)</b>          | 5                    | 6                        | Design all screens, flows, and onboarding experience              |
| 4  | <b>Dev Environment Setup</b>                           | 1                    | 1                        | Configure Flutter, Spring Boot, SQL Server, GitHub                |
| 5  | <b>User Management Module</b>                          | 8                    | 5                        | Auth, 2FA, profile, privacy controls — critical path              |
| 6  | <b>Panoramic Content Module</b>                        | 10                   | 7                        | Capture, stitching, viewer, upload, filters — most complex module |
| 7  | <b>Virtual Gallery &amp; Tour Module</b>               | 8                    | 5                        | Gallery CRUD, virtual tours, scene linking                        |
| 8  | <b>Social Interaction Module</b>                       | 8                    | 5                        | Likes, comments, follow, messaging, stories                       |
| 9  | <b>Analytics &amp; Insights Module</b>                 | 4                    | 3                        | Engagement metrics, reach stats, growth tracking                  |
| 10 | <b>Notification Module</b>                             | 3                    | 2                        | FCM push notifications, preferences, alerts                       |
| 11 | <b>Admin Panel Module</b>                              | 6                    | 4                        | User management, moderation, system config — critical path        |
| 12 | <b>Integration of All Modules</b>                      | 3                    | 3                        | Connect all 7 modules into one working system                     |

|    |                                   |                |                         |                                                 |
|----|-----------------------------------|----------------|-------------------------|-------------------------------------------------|
| 13 | QA Testing & Bug Fixing           | 5              | 5                       | Unit, integration, UAT testing; bug resolution  |
| 14 | Deployment & App Store Submission | 2              | 2                       | Build APK/IPA, submit to stores, cloud deploy   |
| 15 | Documentation & Project Closeout  | 4              | 3                       | User guide, technical docs, academic submission |
|    | TOTAL                             | 76 person-days | 10 weeks (50 cal. days) | Academic project with solo developer            |

Gantt Chart — 10-Week Project Schedule

The Gantt chart below shows all 15 activities plotted across the 10-week project timeline. Dark red bars represent critical path activities — any delay in these will delay the entire project. Light red bars represent non-critical activities that have some scheduling flexibility.

Dark red bar = Critical path activity | Light red bar = Non-critical activity

| # | Activity                        | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | W9 | W10 | Effort (days) | Resource             |
|---|---------------------------------|----|----|----|----|----|----|----|----|----|-----|---------------|----------------------|
| 1 | Requirement Analysis            |    |    |    |    |    |    |    |    |    |     | 5             | Rabia Khalid         |
| 2 | System Architecture & DB Design |    |    |    |    |    |    |    |    |    |     | 4             | Rabia Khalid         |
| 3 | UI/UX Design & Prototyping      |    |    |    |    |    |    |    |    |    |     | 5             | Rabia Khalid (Figma) |
| 4 | Dev Environment Setup           |    |    |    |    |    |    |    |    |    |     | 1             | Rabia Khalid         |
| 5 | User Management Module          |    |    |    |    |    |    |    |    |    |     | 8             | Rabia Khalid         |
| 6 | Panoramic Content Module        |    |    |    |    |    |    |    |    |    |     | 10            | Rabia Khalid         |
| 7 | Virtual Gallery & Tour Module   |    |    |    |    |    |    |    |    |    |     | 8             | Rabia Khalid         |
| 8 | Social Interaction Module       |    |    |    |    |    |    |    |    |    |     | 8             | Rabia Khalid         |
| 9 | Analytics Module                |    |    |    |    |    |    |    |    |    |     | 4             | Rabia Khalid         |

| #  | Activity                           | W1 | W2 | W3 | W4 | W5 | W6 | W7 | W8 | W9 | W10 | Effort (days) | Resource          |
|----|------------------------------------|----|----|----|----|----|----|----|----|----|-----|---------------|-------------------|
| 10 | Notification Module                |    |    |    |    |    |    |    |    |    |     | 3             | Rabia Khalid      |
| 11 | <b>Admin Panel Module</b>          |    |    |    |    |    |    |    |    |    |     | 6             | Rabia Khalid      |
| 12 | <b>Integration of All Modules</b>  |    |    |    |    |    |    |    |    |    |     | 3             | Rabia Khalid      |
| 13 | <b>QA Testing &amp; Bug Fixing</b> |    |    |    |    |    |    |    |    |    |     | 5             | Rabia + Peer Team |
| 14 | <b>Deployment &amp; App Store</b>  |    |    |    |    |    |    |    |    |    |     | 2             | Rabia Khalid      |
| 15 | Documentation & Close              |    |    |    |    |    |    |    |    |    |     | 3             | Rabia Khalid      |

Critical Path: Activity 1 → Activity 2 → Activity 5 → Activity 6 → Activity 7 → Activity 11 → Activity 12 → Activity 13 → Activity 14. This chain has zero float — no delay is permissible on any of these activities.

## Resource Schedule

The resource schedule maps each activity to the person responsible and the tools required. Since 3DPanoSphere is a solo academic project, all development work is handled by Rabia Khalid, with peer review support during testing.

| Activity                         | Responsible          | Tools / Resources                             | Availability           |
|----------------------------------|----------------------|-----------------------------------------------|------------------------|
| <b>Requirements &amp; SRS</b>    | Rabia Khalid         | MS Word, previous SRS templates               | Full-time (Weeks 1–2)  |
| <b>System Architecture</b>       | Rabia Khalid         | Draw.io / Lucidchart, MS Word                 | Full-time (Week 2)     |
| <b>UI/UX Design</b>              | Rabia Khalid         | Figma (free tier)                             | Full-time (Weeks 3–4)  |
| <b>Dev Environment</b>           | Rabia Khalid         | Flutter SDK, Spring Boot, VS Code, SQL Server | Full-time (Week 3)     |
| <b>User Management Module</b>    | Rabia Khalid         | Flutter, Spring Boot, JWT library, FCM        | Full-time (Weeks 4–5)  |
| <b>Panoramic Content Module</b>  | Rabia Khalid         | Flutter panorama library, AWS S3, camera API  | Full-time (Weeks 5–6)  |
| <b>Gallery &amp; Tour Module</b> | Rabia Khalid         | Flutter, Spring Boot REST API, SQL Server     | Full-time (Weeks 6–7)  |
| <b>Social Interaction Module</b> | Rabia Khalid         | WebSocket / Firebase Realtime DB, Flutter     | Full-time (Weeks 6–7)  |
| <b>Analytics Module</b>          | Rabia Khalid         | SQL analytics queries, Flutter charts library | Full-time (Week 7)     |
| <b>Notification Module</b>       | Rabia Khalid         | Firebase Cloud Messaging (FCM), Flutter       | Part-time (Week 5)     |
| <b>Admin Panel Module</b>        | Rabia Khalid         | Spring Boot, Thymeleaf / REST API, SQL Server | Full-time (Weeks 7–8)  |
| <b>Integration</b>               | Rabia Khalid         | All modules, Postman, GitHub                  | Full-time (Week 8)     |
| <b>QA Testing</b>                | Rabia Khalid + Peers | Android Studio, physical devices, Postman     | Full-time (Week 9)     |
| <b>Deployment</b>                | Rabia Khalid         | Google Play Console, Apple Developer account  | Full-time (Week 10)    |
| <b>Documentation</b>             | Rabia Khalid         | MS Word, GitHub README, Figma export          | Part-time (Weeks 9–10) |

## Cost Estimate

Cost estimation for 3DPanoSphere was performed at two levels: the academic project cost (what it actually costs to build as a university project) and the commercial cost (what it would cost if developed as a full professional product with a paid team). Both are important for Step 6.

### Academic Project Cost

This is the actual budget required to complete 3DPanoSphere as an academic project. It was calculated in Lab 03 and covers all real expenses including cloud infrastructure, app store fees, and a contingency reserve.

| Category                        | Item                                            | Cost (PKR) | Cost (USD) | Notes                             |
|---------------------------------|-------------------------------------------------|------------|------------|-----------------------------------|
| <b>Development</b>              | Full-stack development (5 months × PKR 62,500)  | 312,500    | \$1,116    | Solo developer — academic rate    |
| <b>Development</b>              | UI/UX Design (5 months × PKR 15,000)            | 75,000     | \$268      | Figma used for prototyping        |
| <b>Development</b>              | Integration & Testing (5 months × PKR 10,000)   | 50,000     | \$179      | Peer review + QA support          |
| <b>Testing &amp; Deployment</b> | Device testing (Android/iOS)                    | 37,500     | \$134      | University lab + personal devices |
| <b>Testing &amp; Deployment</b> | App Store fees (Google Play + Apple)            | 20,000     | \$71       | One-time Google + annual Apple    |
| <b>Cloud Storage</b>            | Panoramic media storage (6 months)              | 125,000    | \$447      | AWS S3 / Firebase — 500GB         |
| <b>Cloud Storage</b>            | Backend API hosting (6 months)                  | 62,500     | \$223      | DigitalOcean / AWS basic VPS      |
| <b>Infrastructure</b>           | Network setup                                   | 20,000     | \$71       | University campus Wi-Fi           |
| <b>Infrastructure</b>           | Optional local test server                      | 30,000     | \$107      | Internal backup and testing       |
| <b>Software</b>                 | Paid APIs & miscellaneous licenses              | 20,000     | \$71       | Third-party API costs             |
| <b>Marketing</b>                | Social media campaigns & branding               | 50,000     | \$179      | Promotional launch content        |
| <b>Maintenance</b>              | Bug fixes & user support (3 months post-launch) | 50,000     | \$179      | Post-deployment support           |

|                          |                          |            |          |                              |
|--------------------------|--------------------------|------------|----------|------------------------------|
| <b>Contingency (15%)</b> | Unexpected costs reserve | 138,375    | \$494    | Cloud overrun, device issues |
| <b>TOTAL</b>             |                          | ~1,085,000 | ~\$3,860 | Full academic project budget |

Cost-Benefit Analysis

A cost-benefit analysis was performed in Lab 04 to evaluate whether the investment in 3DPanoSphere is financially justified. The analysis used the academic project cost of PKR 742,000 (the development-only portion before contingency).

| Annual Benefit Source                     | Amount (PKR) | Description                                                        |
|-------------------------------------------|--------------|--------------------------------------------------------------------|
| <i>Platform Advertising</i>               | 200,000      | Companies pay to show ads on the platform                          |
| <i>Creator Premium Features</i>           | 150,000      | Creators pay for extra storage and analytics                       |
| <i>Tourism Business Promotions</i>        | 180,000      | Tourism companies pay to promote their locations via virtual tours |
| <i>Sponsored Homepage Posts</i>           | 120,000      | Users pay to feature their posts on the homepage                   |
| <i>Travel Agency &amp; Hotel Partners</i> | 250,000      | Hotels and travel agencies promote via virtual tours               |
| <i>Total Annual Benefits</i>              | 900,000      | Combined projected first-year revenue                              |

| Financial Metric                | Formula                          | Calculation               | Result                  |
|---------------------------------|----------------------------------|---------------------------|-------------------------|
| <i>Net Benefit</i>              | Annual Benefits – Total Cost     | 900,000 – 742,000         | PKR 158,000             |
| <i>BCR (Benefit-Cost Ratio)</i> | Annual Benefits ÷ Total Cost     | 900,000 ÷ 742,000         | 1.21 (> 1.0 = viable)   |
| <i>Payback Period</i>           | Total Cost ÷ Annual Benefits     | 742,000 ÷ 900,000         | 0.82 years (~10 months) |
| <i>ROI</i>                      | (Net Benefit ÷ Total Cost) × 100 | (158,000 ÷ 742,000) × 100 | 21.29%                  |

The cost-benefit analysis confirms that 3DPanoSphere is financially viable. A BCR of 1.21 means every PKR 1 invested returns PKR 1.21 in benefits. The payback period of 0.82 years (~10 months) means the project recovers its full investment in under one year. The ROI of 21.29% represents a healthy return on investment.

Critical Path Analysis

Critical path analysis identifies the longest sequence of dependent activities in the project. Any delay on the critical path directly delays the final delivery date. Non-critical activities have 'float' — some scheduling flexibility before they become a problem.

Critical Path Identification

The critical path for 3DPanoSphere was identified by tracing the longest dependency chain from start to finish. The chain runs through the activities with no scheduling flexibility:

**Critical Path: A1 → A2 → A5 → A6 → A7 → A11 → A12 → A13 → A14**

| Activity | Name                       | Duration | Why It's on the Critical Path                                                                 |
|----------|----------------------------|----------|-----------------------------------------------------------------------------------------------|
| A1       | Requirement Analysis       | 5 days   | Everything depends on requirements being finalized — no activity can start until this is done |
| A2       | System Architecture & DB   | 5 days   | Development of all modules depends on the architecture and database schema being complete     |
| A5       | User Management Module     | 5 days   | All other modules (panoramic, social, admin) depend on authentication being in place          |
| A6       | Panoramic Content Module   | 7 days   | The most complex and time-consuming module; gallery and social features depend on it          |
| A7       | Virtual Gallery & Tour     | 5 days   | Depends on panoramic content; analytics also depends on this being done                       |
| A11      | Admin Panel Module         | 4 days   | Depends on all user modules; must be complete before integration can begin                    |
| A12      | Integration of All Modules | 3 days   | Cannot start until all 7 modules are individually complete                                    |
| A13      | QA Testing & Bug Fixing    | 5 days   | Testing must cover all integrated features; cannot start until integration is done            |
| A14      | Deployment & App Store     | 2 days   | Final step — deployment only happens once testing passes                                      |

**Total Critical Path Duration: 41 calendar days (approximately 8.5 weeks)**

Activities with Float (Non-Critical)

| Activity | Name                  | Float Available | Explanation                                                                             |
|----------|-----------------------|-----------------|-----------------------------------------------------------------------------------------|
| A3       | UI/UX Design          | 2 days          | Can overlap partially with architecture; has some flexibility before dev setup needs it |
| A4       | Dev Environment Setup | 1 day           | Can be done in parallel with late stages of architecture design                         |

|            |                            |          |                                                                                        |
|------------|----------------------------|----------|----------------------------------------------------------------------------------------|
| <i>A8</i>  | <i>Social Interaction</i>  | 0–1 days | Parallel with Gallery module; must be done before integration — very little float      |
| <i>A9</i>  | <i>Analytics Module</i>    | 2 days   | Depends on social + gallery; can start slightly later without delaying integration     |
| <i>A10</i> | <i>Notification Module</i> | 3 days   | Only requires User Module to be done; can be completed in parallel with content module |
| <i>A15</i> | <i>Documentation</i>       | 3 days   | Can be done in parallel with testing and deployment phases                             |

## Step 7: Identify Risks & Risk Management

### Risk Management Overview

Risk management is a critical part of software project management. A risk is an uncertain event that, if it occurs, could have a negative impact on the project's objectives — its schedule, cost, quality, or scope. Step 7 of the MITP framework requires identifying all significant risks, assessing their probability and impact, classifying them, and defining a clear response plan for each.

For 3DPanoSphere, 26 risks were identified in the Risk Information Sheet (RIS) — 13 generic risks that apply to most software projects, and 13 product-specific risks that are unique to a 360° panoramic virtual tour platform.

### Risk Categories

All risks are classified into four categories based on the nature of their source. Understanding the category helps in identifying the right team member to own the risk and the right type of response.

| Code      | Category                   | Risk Count | Risks Included                           | Description                                                                                                        |
|-----------|----------------------------|------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <i>PS</i> | <i>Project Size Risks</i>  | 7          | R1, R2, R3, R14, R15, R21, R23           | Risks related to inaccurate size estimation, unclear requirements, scope changes, and market adoption challenges   |
| <i>ST</i> | <i>Staff / Team Risks</i>  | 3          | R4, R5, R6                               | Risks related to team experience, coordination, availability, and knowledge gaps                                   |
| <i>DE</i> | <i>Development Risks</i>   | 9          | R7, R8, R9, R10, R16, R17, R18, R19, R25 | Risks arising from technical implementation challenges, tool unfamiliarity, integration issues, and UI/UX problems |
| <i>EV</i> | <i>Environmental Risks</i> | 7          | R11, R12, R13, R20, R22, R24, R26        | Risks related to infrastructure, performance, cloud services, security, privacy, and external dependencies         |



Risk Response Strategies

Four strategies are available for responding to an identified risk. The choice depends on the risk level — Critical and High risks cannot simply be accepted.

| Strategy        | Definition                                                                                      | When to Apply to 3DPanoSphere                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <i>Avoid</i>    | Eliminate the source of the risk entirely — change the plan so the risk can no longer occur     | Used for Critical risks where the impact would be project-ending (e.g., security breach, total system failure)  |
| <i>Mitigate</i> | Reduce the probability or impact of the risk through preventive actions taken before it occurs  | Used for High and Medium risks — implement safeguards, prototypes, testing protocols, and coding standards      |
| <i>Transfer</i> | Shift the financial or operational burden of the risk to a third party (insurance, outsourcing) | Used when cloud provider or external service carries the risk (e.g., AWS S3 downtime, FCM notification failure) |
| <i>Accept</i>   | Acknowledge the risk exists but take no proactive action — monitor and prepare a contingency    | Used for Low and Very Low risks where the cost of mitigation exceeds the potential impact                       |

Risk Assessment Matrix

The Risk Assessment Matrix plots all 26 risks based on their probability (horizontal axis: Low / Medium / High) and impact (vertical axis: Catastrophic → Negligible). Each cell shows the combined risk level and which risks fall within it. This visual tool helps prioritise which risks need immediate attention versus which can be monitored passively.

Impact Scale (from RIS document): 1 = Catastrophic (worst), 2 = Critical, 3 = Moderate, 4 = Marginal, 5 = Negligible (least severe)

Probability Bands: Low = below 35%, Medium = 35% to 60%, High = above 60%

| Impact \ Probability | Low Probability<br>(< 35%) | Medium Probability<br>(35–60%)   | High Probability<br>(> 60%) |
|----------------------|----------------------------|----------------------------------|-----------------------------|
| 1 — Catastrophic     | MEDIUM<br>R12, R13, R24    | CRITICAL<br>—                    | CRITICAL<br>—               |
| 2 — Critical         | LOW<br>R6, R11, R20, R26   | HIGH<br>R2, R4, R5, R8, R17, R23 | CRITICAL<br>—               |
| 3 — Moderate         | LOW<br>R19, R25            | HIGH<br>—                        | HIGH<br>R1, R14             |

| Impact \ Probability | Low Probability<br>(< 35%) | Medium Probability<br>(35–60%)    | High Probability<br>(> 60%) |
|----------------------|----------------------------|-----------------------------------|-----------------------------|
|                      |                            | R3, R7, R9, R10, R16,<br>R18, R22 |                             |
| 4 — Marginal         | LOW<br>—                   | MEDIUM<br>R15, R21                | HIGH<br>—                   |
| 5 — Negligible       | VERY LOW<br>—              | LOW<br>—                          | LOW<br>—                    |

Matrix Colour Legend

| Level    | Recommended Action                              |
|----------|-------------------------------------------------|
| CRITICAL | Avoid immediately — escalate to supervisor      |
| HIGH     | Urgent mitigation plan required this sprint     |
| MEDIUM   | Mitigate or transfer — monitor weekly           |
| LOW      | Accept with monitoring — review monthly         |
| VERY LOW | Accept — no action needed unless status changes |

Key Observations from the Matrix

The matrix reveals that 3DPanoSphere has no Critical-level risks in the high-probability column the most dangerous zone. The highest-concern risks sit in the Medium-probability, Critical-impact and High-probability, Moderate-impact zones. R2 (unclear requirements), R14 (storage costs), and R1 (size estimation) are the top-priority risks that require active mitigation plans this sprint.

Most environmental risks (R12, R13, R24, R26) fall in the Low or Very Low zone despite their Catastrophic impact rating, because their probability is low (20–25%). This means standard security practices and backup procedures are sufficient they do not require emergency-level attention unless triggered.

Generic Risk Register (R1 – R13)

Generic risks are risks that apply to virtually all software development projects, regardless of the specific application being built. They cover project sizing, team dynamics, development processes, and environmental factors. For 3DPanoSphere, 13 generic risks were identified in the RIS document.

| ID  | Risk Description                         | Cat. | Prob % | Impact           | Level  | Response Strategy               | RMMM Action                                           |
|-----|------------------------------------------|------|--------|------------------|--------|---------------------------------|-------------------------------------------------------|
| R1  | Project size estimation is inaccurate    | PS   | 70%    | 3 – Moderate     | HIGH   | Mitigate — action plan required | Use COCOMO II + bottom-up; divide into modules        |
| R2  | Requirements not clearly defined         | PS   | 60%    | 2 – Critical     | HIGH   | Mitigate — action plan required | Finalize and freeze SRS early in sprint 1             |
| R3  | Frequent requirement changes             | PS   | 50%    | 3 – Moderate     | HIGH   | Mitigate — action plan required | Freeze scope after design; use change control         |
| R4  | Lack of team experience in technologies  | ST   | 55%    | 2 – Critical     | HIGH   | Mitigate — action plan required | Train in Flutter & Spring Boot before dev             |
| R5  | Low user adoption due to competition     | PS   | 50%    | 2 – Critical     | HIGH   | Mitigate — action plan required | Highlight unique features; strengthen marketing       |
| R6  | Key team member becomes unavailable      | ST   | 30%    | 2 – Critical     | MEDIUM | Mitigate / Transfer             | Maintain backup resources and documentation           |
| R7  | Unfamiliar tools slow development        | DE   | 45%    | 3 – Moderate     | MEDIUM | Mitigate / Transfer             | Learn tools in Sprint 1; prototype before full dev    |
| R8  | Integration issues between modules       | DE   | 40%    | 2 – Critical     | HIGH   | Mitigate — action plan required | Modular design; early continuous integration testing  |
| R9  | Poor UI design affects usability         | DE   | 35%    | 3 – Moderate     | MEDIUM | Mitigate / Transfer             | Usability testing with real users; iterate prototypes |
| R10 | Software bugs increase during dev        | DE   | 50%    | 3 – Moderate     | HIGH   | Mitigate — action plan required | Regular unit and integration testing; bug tracking    |
| R11 | Performance issues under concurrent load | EV   | 30%    | 2 – Critical     | MEDIUM | Mitigate / Transfer             | Optimize code; load testing before deployment         |
| R12 | Data loss due to cloud failure           | EV   | 20%    | 1 – Catastrophic | MEDIUM | Mitigate / Transfer             | Automated backups; AWS S3 versioning enabled          |
| R13 | Security vulnerabilities                 | EV   | 25%    | 1 – Catastrophic | MEDIUM | Mitigate / Transfer             | JWT auth, 2FA, HTTPS encryption, OWASP practices      |

### Product-Specific Risk Register (R14 – R26)

Product-specific risks are unique to the nature of 3DPanoSphere as a 360° panoramic virtual tour sharing platform. They relate to the core features of the system panoramic image handling, viewer performance, hotspot functionality, virtual tour navigation, real-time messaging, and the star rating system. These risks were identified specifically from the technical and functional requirements of this project.

| ID  | Risk Description                               | Cat. | Prob % | Impact           | Level  | Response Strategy               | RMMM Action                                          |
|-----|------------------------------------------------|------|--------|------------------|--------|---------------------------------|------------------------------------------------------|
| R14 | Large panoramic images cause budget overrun    | PS   | 65%    | 3 – Moderate     | HIGH   | Mitigate — action plan required | Image compression 50–70%; per-user storage limits    |
| R15 | Users upload low-quality panoramas             | PS   | 50%    | 4 – Marginal     | MEDIUM | Mitigate / Transfer             | Auto quality checks; provide upload guidelines       |
| R16 | Difficulty stitching panorama images           | DE   | 45%    | 3 – Moderate     | MEDIUM | Mitigate / Transfer             | Use OpenCV stitching libraries; test diverse images  |
| R17 | Viewer doesn't work smoothly on all devices    | DE   | 40%    | 2 – Critical     | HIGH   | Mitigate — action plan required | Optimize viewer for low-end devices; minimum specs   |
| R18 | Interactive hotspots not functioning correctly | DE   | 35%    | 3 – Moderate     | MEDIUM | Mitigate / Transfer             | Test hotspot functionality across devices            |
| R19 | Virtual tour navigation becomes confusing      | DE   | 30%    | 3 – Moderate     | LOW    | Accept with monitoring          | Simple navigation UI; usability testing on tours     |
| R20 | Real-time messaging increases backend load     | EV   | 30%    | 2 – Critical     | MEDIUM | Mitigate / Transfer             | Scalable WebSocket / Firebase; load testing          |
| R21 | Star rating misused through fake ratings       | PS   | 35%    | 4 – Marginal     | LOW    | Accept with monitoring          | Limit one rating per verified user; monitor activity |
| R22 | Slow internet affects panorama loading         | EV   | 40%    | 3 – Moderate     | MEDIUM | Mitigate / Transfer             | Progressive loading and caching; compress media      |
| R23 | Low user adoption due to competition           | PS   | 50%    | 2 – Critical     | HIGH   | Mitigate — action plan required | Highlight unique features; strengthen marketing      |
| R24 | Data privacy issues in user content            | EV   | 25%    | 1 – Catastrophic | MEDIUM | Mitigate / Transfer             | Privacy controls; user consent; GDPR compliance      |
| R25 | Difficulty linking panoramas into tour         | DE   | 30%    | 3 – Moderate     | LOW    | Accept with monitoring          | Structured tour-linking module; test with content    |

| ID  | Risk Description                | Cat. | Prob % | Impact       | Level | Response Strategy      | RMMM Action                                         |
|-----|---------------------------------|------|--------|--------------|-------|------------------------|-----------------------------------------------------|
| R26 | App crashes due to memory usage | EV   | 20%    | 2 – Critical | LOW   | Accept with monitoring | Optimize memory; lazy-load; test low-memory devices |

## Step 8: Review & Publicize the Plan

### Complete Plan Summary — Steps 1 To 8

Before reviewing and publishing the plan, it is essential to confirm that all eight planning steps have been completed. This section provides a summary of every step — what was done, what was produced, and what the key results were. This serves as the executive overview of the entire project plan.

| Step          | Name                                            | Output Document                                              | Key Results                                                                                                          |
|---------------|-------------------------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Step 1</b> | <b>Select Project &amp; Determine Scope</b>     | Project Charter + High-Level Feasibility Report              | Option 3 selected — Full 3D platform; BCR 1.21; net benefit PKR 158,000                                              |
| <b>Step 2</b> | <b>Identify Project Scope &amp; Objectives</b>  | Project Definition Document (PDD)                            | 16 deliverables; 8 SMART objectives; 9 stakeholders; communication plan; sign-off requirements                       |
| <b>Step 3</b> | <b>Identify Project Infrastructure</b>          | Infrastructure Inventory & Dependency Map                    | Flutter + Spring Boot + SQL Server + AWS; IEEE 830 + Agile + OWASP standards; solo dev reporting to supervisor       |
| <b>Step 4</b> | <b>Analyse Project Characteristics</b>          | Project Characteristics Report & Methodology Choice          | Well-defined, data-intensive, real-time, interactive, mobile-first, cloud-based app; Agile (2-week sprints) selected |
| <b>Step 5</b> | <b>Identify Products &amp; Activities (WBS)</b> | Work Breakdown Structure — 5 levels, 3 PlantUML diagrams     | 7 deliverable modules; 15 activities; dependency map; precedence sequence; Process/Product/Hybrid WBS                |
| <b>Step 6</b> | <b>Estimate Effort, Duration &amp; Schedule</b> | Gantt Chart, Resource Schedule, Cost Estimate, Critical Path | 76 person-days; 10 weeks; PKR 1,085,000 academic budget; \$54,000 COCOMO II; critical path 41 days                   |
| <b>Step 7</b> | <b>Identify Risks &amp; Risk Management</b>     | Risk Register + Risk Assessment Matrix + RMMM Plan           | 26 risks (R1–R26); 5×3 risk matrix; RMMM plan with mitigation, monitoring & contingency for each risk                |
| <b>Step 8</b> | <b>Review &amp; Publicise the Plan</b>          | Baseline Project Plan Document (This Document)               | 6-item review checklist confirmed; 4 publication deliverables distributed; formal sign-off by supervisor             |

## Plan Review Checklist

The Plan Review Checklist verifies that every essential component of a sound project plan is in place before the plan is published and execution begins. All six items must be confirmed before the supervisor signs off. Each item references the specific step where it was completed and the evidence that confirms it.

| Checklist Item                             | Step Covered | Status               | Evidence / Where to Find It                                                                                                                                               |
|--------------------------------------------|--------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>All steps completed and documented?</i> | Steps 1–7    | ✓<br><b>COMPLETE</b> | Steps 1–7 each have a dedicated .docx document covering all activities, outputs, and sign-off sections                                                                    |
| <i>Risks identified and mitigated?</i>     | Step 7       | ✓<br><b>COMPLETE</b> | 26 risks identified in the RIS (13 generic + 13 product-specific); full RMMM plan documented with mitigation, monitoring, and contingency for each                        |
| <i>Resources allocated realistically?</i>  | Step 6       | ✓<br><b>COMPLETE</b> | Resource schedule assigns Rabia Khalid to all 15 activities; Peer Review Team assigned to QA (Week 9); all tools and cloud services listed with costs                     |
| <i>Deadlines and milestones set?</i>       | Steps 5 & 6  | ✓<br><b>COMPLETE</b> | 15 activities with individual durations scheduled across 10 weeks; Gantt chart produced; critical path (41 days) identified; deadline June 7, 2026 confirmed              |
| <i>All stakeholders reviewed the plan?</i> | Steps 2 & 8  | ✓<br><b>COMPLETE</b> | 9 stakeholders identified in Step 2; communication plan defined; sign-off requirements listed; supervisor Ma'am Aneela Abbas is the formal approver                       |
| <i>Quality measures defined?</i>           | Steps 2 & 6  | ✓<br><b>COMPLETE</b> | KPIs defined: $\geq 85\%$ usability, $\geq 4/5$ satisfaction, $< 3s$ response time, $< 3\%$ error rate; QA testing phase (Week 9) with unit, integration, and UAT testing |

**All 6 checklist items are confirmed COMPLETE. The plan is ready for supervisor review and formal sign-off.**

## Publish & Communicate The Plan

Publishing the plan means making it formally available to all key stakeholders in an agreed format. For 3DPanoSphere, four items are published as part of the Step 8 deliverable. Each item serves a specific purpose for a specific audience.

### Publication & Distribution Plan

| Publication Item                       | Description                                                                                                         | Distributed To                                                            | Format            | Status                     |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------|----------------------------|
| <b><i>Project Plan Document</i></b>    | Complete baseline plan covering all 8 steps — scope, objectives, infrastructure, WBS, schedule, risks, and review   | Ma'am Aneela Abbas (Supervisor), University CS Department, Project File   | .docx + PDF       | Ready for distribution     |
| <b><i>Gantt Chart / Schedule</i></b>   | 10-week visual project timeline with all 15 activities, durations, critical path (dark red), and sprint assignments | Ma'am Aneela Abbas, Peer Review Team (for testing schedule), Project file | .docx (landscape) | Ready for distribution     |
| <b><i>Risk Register</i></b>            | Complete 26-risk register with probability, impact, risk level (colour-coded), response strategy, and RMMM plan     | Ma'am Aneela Abbas, Developer (self), Peer QA Team                        | .docx table       | Ready for distribution     |
| <b><i>Sign-off Meeting Minutes</i></b> | Formal record of the sign-off meeting agenda, decisions made, approvals given, and any conditions noted             | Ma'am Aneela Abbas, University CS Department, Project Record              | .docx             | Pending supervisor meeting |

### Stakeholder Communication Summary

Each stakeholder group receives specific information based on their role and level of interest in the project. The communication approach ensures that no stakeholder is left uninformed about the project plan.

| Stakeholder                                   | What They Receive                                                  | Purpose                                         | Channel                    |
|-----------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------|----------------------------|
| <b><i>Ma'am Aneela Abbas (Supervisor)</i></b> | Complete Step 8 document + all Step 1–7 documents + sign-off sheet | Formal review and approval; academic assessment | Printed + email submission |



| Stakeholder                        | What They Receive                                                     | Purpose                                               | Channel                        |
|------------------------------------|-----------------------------------------------------------------------|-------------------------------------------------------|--------------------------------|
| <b>University CS Department</b>    | Final submitted project plan package                                  | Academic evaluation against BSSE curriculum           | University submission portal   |
| <b>Peer Review / QA Team</b>       | Gantt chart (testing weeks highlighted) + QA testing plan from Step 6 | Coordinate testing schedule and responsibilities      | WhatsApp / lab session         |
| <b>Beta Testers (End Users)</b>    | Usability testing schedule and instructions                           | Participate in UAT testing during Week 9              | Email / direct contact         |
| <b>Rabia Khalid (Developer/PM)</b> | Complete baseline plan as personal working reference                  | Day-to-day execution guide; sprint planning reference | Personal project file / GitHub |

### Sign-Off Meeting

The sign-off meeting is the formal event at which all key stakeholders review the completed project plan and grant approval to proceed to project execution. For 3DPanoSphere, this is a structured meeting between Rabia Khalid (Project Manager) and Ma'am Aneela Abbas (Project Supervisor). The meeting formally closes the planning phase and opens the execution phase.

### Sign-Off Meeting Agenda

| # | Agenda Item                               | Duration | Discussion Points                                                                     | Owner                     |
|---|-------------------------------------------|----------|---------------------------------------------------------------------------------------|---------------------------|
| 1 | <b>Welcome &amp; Introduction</b>         | 5 min    | Purpose of the sign-off meeting; confirm all attendees present                        | Rabia Khalid              |
| 2 | <b>Review of Planning Steps 1–7</b>       | 20 min   | Walk through each step's output document; confirm all items are complete and accurate | Rabia Khalid              |
| 3 | <b>Plan Review Checklist Walk-Through</b> | 10 min   | Confirm all 6 checklist items are satisfied; address any open questions               | Rabia Khalid + Supervisor |
| 4 | <b>Gantt Chart &amp; Schedule Review</b>  | 10 min   | Review 10-week Gantt chart; confirm milestone dates; acknowledge critical path        | Rabia Khalid              |
| 5 | <b>Risk Register Review</b>               | 10 min   | Confirm all 26 risks are documented; review top HIGH risks and their RMMM plans       | Rabia Khalid              |

| #  | Agenda Item                               | Duration | Discussion Points                                                                                   | Owner                     |
|----|-------------------------------------------|----------|-----------------------------------------------------------------------------------------------------|---------------------------|
| 6  | <b>Resource &amp; Budget Confirmation</b> | 5 min    | Confirm PKR 1,085,000 budget; resource allocation to all 15 activities                              | Rabia Khalid              |
| 7  | <b>Quality Measures Confirmation</b>      | 5 min    | Confirm KPIs and quality gates; agree on UAT process                                                | Rabia Khalid + Supervisor |
| 8  | <b>Change Control Agreement</b>           | 5 min    | Agree on formal change control process for scope, schedule, budget, and risk changes                | Both parties              |
| 9  | <b>Baseline Confirmation</b>              | 5 min    | Formally agree that this plan is the project baseline                                               | Both parties              |
| 10 | <b>Sign-Off Approval</b>                  | 5 min    | Supervisor signs the Plan Review Checklist and Baseline document; project execution formally begins | Ma'am Aneela Abbas        |

### Formal Sign-Off Signatures

The signatures below formally confirm that the project plan has been reviewed, all checklist items are satisfied, and the plan is approved as the project baseline. After signing, any changes to the plan require formal change control.

| Stakeholder / Role                     | Name                      | Authority Level | Signature | Date of Approval |
|----------------------------------------|---------------------------|-----------------|-----------|------------------|
| <b>Project Manager &amp; Developer</b> | <b>Rabia Khalid</b>       | Prepared By     | _____     | _____            |
| <b>Project Supervisor</b>              | <b>Ma'am Aneela Abbas</b> | Final Approver  | _____     | _____            |
| <b>Academic Oversight</b>              | <b>CS Department, UoC</b> | Evaluator       | _____     | _____            |

## Quality Measures

Quality measures define how the team will know that the project is on track and that the final product meets the required standards. These measures were defined in Steps 2 and 6 and are confirmed here as part of the baseline plan. They form the quality gates that must be passed before the project can proceed from testing to deployment.

| Quality Area              | Measure / KPI                                 | Target               | Measurement Method                               | Responsible              |
|---------------------------|-----------------------------------------------|----------------------|--------------------------------------------------|--------------------------|
| <b>System Usability</b>   | Task completion rate without assistance       | $\geq 85\%$          | Usability test with 100 users (Week 9)           | Rabia Khalid + Peer Team |
| <b>User Satisfaction</b>  | Average in-app feedback rating                | $\geq 4 / 5$         | In-app survey after UAT                          | Rabia Khalid             |
| <b>System Performance</b> | API response time under normal load           | $< 3$ seconds        | Postman + performance testing tools              | Rabia Khalid             |
| <b>Media Loading</b>      | 360° panorama load time                       | $< 5$ seconds        | Device testing on mid-range Android device       | Rabia Khalid             |
| <b>Error Rate</b>         | Operation failure rate across all modules     | $< 3\%$              | System logs + automated test reports             | Rabia Khalid             |
| <b>Concurrent Users</b>   | Load handling capacity                        | $\geq 500$ users     | Load testing before deployment                   | Rabia Khalid             |
| <b>Code Quality</b>       | No critical bugs in production                | 0 P1 bugs            | Unit + integration testing; GitHub issue tracker | Rabia Khalid + Peer QA   |
| <b>Security</b>           | Authentication, encryption, input validation  | All OWASP tests pass | Security testing before deployment               | Rabia Khalid             |
| <b>Documentation</b>      | All required documents completed and approved | 100% complete        | Supervisor review and sign-off                   | Rabia Khalid             |
| <b>Schedule Adherence</b> | Activities completed on time                  | $\leq 10\%$ variance | Weekly milestone tracking against Gantt chart    | Rabia Khalid             |

All quality gates listed above must be passed during the QA Testing phase (Week 9) before the deployment activity (Week 10) can begin. Any gate that fails must be resolved and re-tested within the same sprint. If unresolvable, a formal change request must be raised.

## Approved Baseline Plan

The Baseline Plan is the formally agreed version of the project plan at the point of sign-off. It captures all the key planning decisions that were made across Steps 1–8. Any deviation from these values after sign-off requires formal change control.

| Baseline Item             | Approved Value                                            | Change Control Required?                       |
|---------------------------|-----------------------------------------------------------|------------------------------------------------|
| <b>Project Name</b>       | 3DPanoSphere – 3D Virtual Tour Sharing Platform           | No — fixed                                     |
| <b>Total Deliverables</b> | 16 modules and documents across 8 planning steps          | Yes — scope change process                     |
| <b>Project Duration</b>   | 10 weeks (November 22, 2025 – June 7, 2026)               | Yes — schedule change process                  |
| <b>Total Effort</b>       | 76 person-days (bottom-up) / 45 person-months (COCOMO II) | Yes — if scope changes                         |
| <b>Academic Budget</b>    | PKR 1,085,000 (~\$3,860 USD)                              | Yes — budget change process                    |
| <b>Team Size</b>          | 1 developer (Rabia Khalid) + Peer QA support              | Yes — resource change                          |
| <b>Development Method</b> | Agile — 5 sprints of 2 weeks each                         | Yes — methodology change process               |
| <b>Critical Path</b>      | A1→A2→A5→A6→A7→A11→A12→A13→A14 (41 days)                  | Yes — any critical path delay triggers re-plan |
| <b>Total Risks Logged</b> | 26 risks (R1–R26); highest level = HIGH                   | Yes — new risks added via risk register update |
| <b>Quality Benchmark</b>  | ≥85% usability, ≥4/5 satisfaction, <3s response           | Yes — quality gate must be maintained          |
| <b>App Store Target</b>   | Google Play Store + Apple App Store                       | No — deployment platforms fixed                |
| <b>Final Submission</b>   | June 7, 2026                                              | Yes — requires supervisor approval to extend   |